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Reliable Causal Estimation under Latent Markov Confounding

Overlap-aware regime Double Machine Learning, with evidence from Indian urban air-quality sensor networks

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the credit requirement of the Minor in Data Science (8 credits), towards the award of the degree of **Bachelor of Technology in Electrical and Electronics Engineering**, School of Electrical Engineering, Manipal Institute of Technology, Manipal (7th semester, academic year 2026-27).

DECLARATION

We hereby declare that this minor project report is the record of the work carried out by us, and that the results, tables, figures and code reported here are those of the project repository, are regenerated from committed artifacts, and have not been submitted elsewhere for the award of any degree or diploma. Prior work is credited in the text and in the references. Large language models were used as assistants for code, derivations, experiment scripting and drafting; every number is produced by a single script from committed result files and the authors reviewed all arguments and take full responsibility for the content (Annexure B).

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ABSTRACT

Observational time series are frequently driven by a persistent state that nobody measures: the boundary-layer regime of an airshed, the liquidity regime of a market, the operating mode of a plant. When that state moves both a treatment and an outcome, adjusting for observed covariates does not identify the effect. The standard remedy is a two-stage recipe: infer the state with a hidden Markov model (HMM), then run regime-aware Double Machine Learning (DML). The recipe carries an implicit reliability rule: trust the estimate when the posterior is confident and the score Jacobian is well conditioned.

This project shows that the rule fails in a specific and checkable way. A generated state does not only add noise to the downstream estimating equation; it changes that equation's geometry, so that the smallest eigenvalue of the generated-state Gram matrix can *grow* as the state estimate gets worse (it grows 70 times in a controlled design while causal error first rises and then falls, and in 82% of learned-posterior runs the generated state looks better conditioned than the true one). We call this *phantom causal resolution*. We derive two exact laws: the population displacement of a coupled regime score under state confusion, which reproduces a 4,900-run factorial experiment to within 2%, and a residual-state bias law, an omitted-variable identity in the posterior's leftover state variance. The laws dictate the estimator: *overlap-aware regime DML* (OR-DML) conditions its nuisances on the posterior rather than weighting them by it, which makes the coupled score exactly Neyman-orthogonal for every posterior, and combines purged block cross-fitting, spectral regularization and HAC inference.

Evidence comes from factorial and learned-posterior synthetic designs, a real-data-calibrated semi-synthetic benchmark on four Indian cities (Delhi, Mumbai, Bengaluru, Kolkata) on a corrected hourly grid, and an audited observational stress test with aligned placebos and robustness values. Posterior-adjusted OR-DML is the most accurate regime-aware estimator in every synthetic frontier cell and in every semi-synthetic city for regime effects; no estimator removes residual-state bias under weak observability, and the report states where each result stops. The work yielded the peer-reviewed manuscript submitted to AISTATS 2027, a verified reproduction pipeline in which every number is generated from committed artifacts, and an honest account of an earlier applied study that an internal audit led us to retire.

Keywords: causal inference, double machine learning, latent Markov models, regime switching, phantom causal resolution, air quality, reproducibility.

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NOTATION AND ABBREVIATIONS

Symbol	Meaning
$S_t \in \{1, \dots, K\}$	latent regime at time t (first-order Markov chain)
Z_t, X_t	regime proxies (meteorology) and observed controls (lagged pollutants, weather)
T_t, Y_t	treatment (NO_2) and outcome ($PM_{2.5}$)
γ_{tk}	HMM posterior probability $P(S_t = k \mid \cdot)$
θ^*_k, θ^*_{ATE}	regime-k effect and stationary average effect
ε_γ	posterior error $\mathbb{E}\lVert\gamma_t - H_t\rVert_1$ against the one-hot true state H_t
$\hat{J}, \hat{S}, \lambda_{\min}$	coupled Gram matrix, score vector, and its smallest eigenvalue
$v_t = \gamma_t(1-\gamma_t)$	posterior variance of the regime indicator (residual state variance)
τ^*	embargo (purge) length in time steps
$b^\star$	robustness value: smallest outcome regime shift (in SD of $\tilde{Y}$) that explains the estimate away

Abbreviation	Expansion	Abbreviation	Expansion
AISTATS	Intl. Conf. on Artificial Intelligence and Statistics	HAC	heteroskedasticity- and autocorrelation-consistent
DML	Double / Debiased Machine Learning	HMM	hidden Markov model
OR-DML	overlap-aware regime DML	EM	expectation–maximization
ATE	average treatment effect	CPCB	Central Pollution Control Board
DGP	data-generating process	ECE / AUC	expected calibration error / area under ROC

CHAPTER 1

Introduction

1.1 Background and motivation

Air pollution is the largest environmental health risk in India, and the question that policy makers ask of it is causal rather than predictive: *if emissions of a precursor fall, by how much will fine particulate matter ($PM_{2.5}$) fall?* A forecasting model, however accurate, does not answer that question. A model that predicts tomorrow's $PM_{2.5}$ from today's NO_2 will happily report a strong positive association even when most of the association is produced by a third factor, the state of the atmosphere, that raises both pollutants together.

The atmosphere is the textbook example of a persistent, unmeasured state. In a continental basin such as Delhi, winter nights alternate between *stagnation* (a shallow inversion traps everything near the ground) and *ventilation* (wind clears the basin). In a coastal city such as Mumbai, sea and land breezes alternate. Each regime lasts hours to days, so the state is autocorrelated, and each regime shifts both the emission-driven part of NO_2 and the background accumulation of $PM_{2.5}$. Adjusting for observed temperature or wind does not remove this confounding if the regime is only imperfectly captured by those variables.

The same structure appears far outside air quality: the liquidity regime of a market that moves both an execution policy and its price impact, the physiological phase of a patient that moves both a treatment and an outcome, the operating mode of a plant. The statistical problem is general. We call it **causal estimation under latent Markov confounding**.

1.2 Problem statement

We observe a time series $(Y_t, T_t, X_t, Z_t)_{t=1}^{N}$ in which a latent first-order Markov regime S_t affects both treatment T_t and outcome Y_t, the regime is observed only through proxies Z_t, and the target is the regime-specific effect of T on Y and its stationary average. The prevailing approach is a two-stage recipe: fit a hidden Markov model to Z to obtain posterior regime probabilities, then feed them into a regime-aware Double Machine Learning (DML) estimator. Practitioners judge the result with a reliability rule that has two parts: *trust the estimate when the posterior is confident and the score Jacobian is well conditioned.*

The project asks three questions.

1. **Is the reliability rule sound?** When the state is *generated* (estimated) rather than observed, does a confident posterior together with a well-conditioned Jacobian certify the causal estimate?
2. **If not, why, and what should the estimator look like?** Can the failure be characterized exactly, and does the characterization dictate a better design?
3. **What can be said about real sensor data?** On Indian megacity sensor networks, what can an honest analysis conclude, and what must it refuse to conclude?

1.3 Objectives

The objectives agreed at the start of the minor project, and their outcome, are summarized below.

Table 1.1. Project objectives and where each is addressed

#	Objective	Outcome	Chapter
O1	Assemble and audit a multi-city hourly pollution and weather dataset	Four cities, 12 pollutant/weather series, time-grid defect found and corrected	6, 8
O2	Characterize how latent-state error propagates into causal error	Two exact laws; phantom causal resolution discovered	3, 5
O3	Design and implement a reliable regime-aware estimator	OR-DML with posterior-adjusted nuisances, purged cross-fitting, spectral regularization, HAC inference	4
O4	Validate against synthetic, semi-synthetic and real data	4,900-run factorial, 900-dataset frontier, 4-city calibrated benchmark, audited stress test	7, 8
O5	Produce a peer-reviewed research submission	AISTATS 2027 manuscript with 8-page main body and appendices A–Q	10
O6	Make every claim reproducible and machine-checked	Single-script number pipeline, verifier, CI	9

1.4 Summary of contributions

- **A failure and its limits.** An exact decomposition of the omitted-regime bias of standard DML, including bias amplification (adding observed controls can make it worse), and a non-identification result when proxies carry no state information.
- **A mechanism.** Exact laws for how state confusion enters coupled and posterior-conditioned estimators. They explain *phantom causal resolution* and the non-monotone error, and why a posterior helps as an adjustment feature but hurts as a nuisance weight.
- **An estimator.** OR-DML: posterior-adjusted nuisances, purged block cross-fitting, spectral regularization, HAC inference. It comes with exact orthogonality, a degradation bound that separates nuisance error from state-recovery error, and asymptotic normality that includes occupation variance.

- **Evidence.** A factorial design verified against the closed form, learned-posterior experiments that remove its circularity, a real-data-calibrated semi-synthetic benchmark on four Indian cities, and an audited stress test with aligned placebos and robustness values.
- **A research artifact.** A reproduction pipeline in which every number in the manuscript is generated from committed files, and a documented audit trail of every claim that changed along the way (Chapter 10).

1.5 Scope and honesty statement

The report separates three kinds of statements: *theorems* (exact, under stated assumptions), *simulation evidence* (known truth, so errors are measurable), and *observational findings* (no known truth, so they are associations, not effects). The Delhi, Mumbai, Bengaluru and Kolkata analyses are in the third group. We do not claim that NO_2 causes $PM_{2.5}$ with a stated magnitude, and we do not claim that OR-DML dominates every competitor on every metric. Where posterior-feature DML beats OR-DML (the pooled ATE in two cities) or where every estimator fails (weak proxy separation) the report says so, as does the manuscript.

1.6 Organization of the report

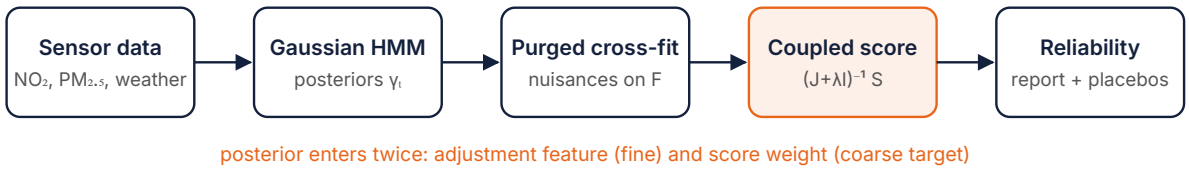


Figure 1.1. Method pipeline followed throughout the report: sensor data are summarized by an HMM, the posterior enters the nuisances as a feature, cross-fitting is purged for temporal dependence, and a regularized coupled score gives regime effects together with a reliability report.

Chapter 2 surveys the literature. Chapter 3 formalizes the model and proves the bias and non-identification results. Chapter 4 presents OR-DML and its implementation. Chapter 5 develops the theory: orthogonality, graceful degradation, the exact laws, and inference. Chapter 6 describes data and experimental design. Chapter 7 reports synthetic and semi-synthetic results. Chapter 8 reports the observational stress test. Chapter 9 documents engineering, reproducibility and verification. Chapter 10 tells the story of the project, including the audit that redirected it. Chapter 11 concludes. The annexures hold team roles, the AI-use statement, code listings, reproduction commands, additional tables, and a map of the manuscript's appendices.

CHAPTER 2

Background and Literature Survey

This chapter places the project in the literature that surrounds it. It is organised from the general to the specific: the potential-outcomes framework (2.1), debiased machine learning (2.2), dependent data (2.3), latent confounders and regime models (2.4), proxies and negative controls (2.5), omitted-variable bias, amplification and overlap (2.6), sensitivity analysis (2.7), and the applied literature on air-quality attribution (2.8). Section 2.9 collects the comparison and states the gap. Every reference cited below was checked against a bibliographic database (Crossref or OpenAlex) for title, authors, venue and year before it was added to the bibliography. Table 2.1 maps the sections to the chapters of this report that use them.

Table 2.1. Where each strand of the survey is used in this report

Section	Strand	Used in
2.1–2.2	Potential outcomes, DML, orthogonal scores	Ch. 3 (problem), Ch. 4 (estimator)
2.3	Dependent-data cross-fitting, mixing, HAC inference	Ch. 4 (purged blocks), Ch. 5 (theory)
2.4	Latent confounders, HMM regimes, stepwise latent-class estimation	Ch. 3, Ch. 5 (plug-in posterior law)
2.5	Proxies and negative controls	Ch. 8 (placebos), Ch. 11 (threats)
2.6	Omitted-variable bias, amplification, overlap	Ch. 5 (bias laws), Ch. 7 (phantom resolution)
2.7	Sensitivity analysis	Ch. 5, Ch. 8 (robustness value $b^{\star}$)
2.8	Air-quality attribution	Ch. 6, Ch. 8 (cities), Ch. 11

2.1 Causal inference from observational data

Potential outcomes and identification. The potential-outcomes view of causal effects goes back to Rubin (1974), and Holland (1986) formalised its logic and its fundamental limitation: only one potential outcome is ever observed per unit. Effects are identified from observational data under *unconfoundedness* (no unmeasured confounding given covariates) and overlap. Rosenbaum and Rubin (1983) showed that conditioning on the propensity score suffices for adjustment, and Hahn (1998) established the semiparametric efficiency bound for average treatment effects that later estimators target. Imbens (2004) reviews the nonparametric estimators that follow from these conditions, and Angrist et al. (1996) give the complementary instrumental-variable route when unconfoundedness is not credible. The graphical alternative expresses the same

assumptions as a diagram, with identification read from the structure of paths (Pearl (1995)); a textbook treatment of the potential-outcomes approach is Imbens and Rubin, 2015.

Continuous treatments and the partially linear model. With a continuous treatment and many confounders, adjustment is done by regression, by weighting, or by a combination. The partially linear model $Y = \theta T + g(X) + U$ with $T = m(X) + V$ is the workhorse: the effect θ is the coefficient on the *residual* of T after removing what the controls explain. This residualisation device is due to Robinson (1988), who obtained $\sqrt{N}$ -consistent estimation of θ with nonparametric nuisance estimates. It is the starting point of the estimator used in this project.

Observational emulation. Hernán and Robins (2016) formalise the idea that an observational analysis should be designed to emulate a hypothetical randomised "target trial" (eligibility, treatment strategy, time zero, follow-up). We borrow the discipline, not the machinery: Chapter 3 states the structural model and the target estimand before any estimator is fitted. Rubin (2008) argues the same point from the design side: the analysis plan should be fixed before outcomes are examined.

2.2 Double / Debiased Machine Learning

Chernozhukov et al. (2018) showed that if the nuisance functions $m(X) = \mathbb{E}[T \mid X]$ and $\mu(X) = \mathbb{E}[Y \mid X]$ are estimated by flexible machine learning, then the residual-on-residual estimator of θ is still $\sqrt{N}$-consistent and asymptotically normal, provided two ingredients are present.

- **Neyman orthogonality.** The estimating equation is insensitive, to first order, to errors in the nuisances. First-order nuisance mistakes then contribute only second-order terms, which are negligible when nuisance rates are $o(N^{-1/4})$.
- **Cross-fitting.** Nuisances are estimated on data not used to evaluate the equation, removing own-observation overfitting bias.

Antecedents. Orthogonal scores are older than the DML label. Newey (1994) derived the influence-function calculus that gives the asymptotic variance of semiparametric estimators; Robins et al. (1994) constructed estimators that stay consistent when either the outcome model or the treatment model is correct, and Bang and Robins (2005) gave the doubly robust formulation that is the template for orthogonal estimating equations. Belloni et al. (2014) and Belloni et al. (2014) showed that post-selection inference for treatment effects with many controls requires exactly this type of orthogonality, and Chernozhukov et al. (2017) previewed the DML construction for treatment effects. Kennedy (2016) gives a readable account of the empirical-process arguments behind cross-fitting, and Hines et al. (2022) explain how efficient influence functions yield orthogonal scores in practice.

Extensions. Automatic debiasing (Chernozhukov, Newey and Singh, 2022) removes the need to derive the correction term by hand, with neural and forest versions in Chernozhukov et al. (2022). Locally robust estimation for general functionals is treated by Chernozhukov et al. (2022). Orthogonal statistical learning (Foster and Syrgkanis, 2023) and quasi-oracle heterogeneous effects (Nie and Wager, 2021) extend the idea to conditional effects; Semenova and Chernozhukov (2021) give debiased inference on conditional average effects, and Kennedy (2023) derives optimal doubly robust estimation of heterogeneous effects. Deep networks are shown to be admissible nuisance learners in Farrell et al. (2021). Tree ensembles targeted at heterogeneity are developed in Athey and Imbens (2016), Wager and Athey (2018) and Athey et al. (2019); meta-learner decompositions are compared in Künzel et al. (2019). DML has also been adapted to difference-in-differences designs (Chang (2020)) and to conditional effects in high dimensions (Fan et al. (2022)).

What is common to these results. They are stated for independent data and *observed* confounders. The two assumptions that this project relaxes, independence and observability of the confounder, are exactly the assumptions on which the guarantees above rest.

2.3 Dependent data: time-series DML

Why independence matters. Time series break the independence assumption. Random K-fold cross-fitting leaks information across neighbouring observations, because training and evaluation folds are correlated. Validity of cross-validation for dependent data is a long-standing concern in forecasting: Stone (1974) introduced cross-validatory assessment for independent data; Racine (2000) showed that for dependent data one must remove a block of observations around the evaluation point (*hv-block* validation); Bergmeir and Benítez (2012) and Bergmeir et al. (2018) document when plain K-fold remains acceptable for autoregressive models and when it does not. In financial machine learning, *purging* and *embargo* of training observations adjacent to the evaluation block are standard remedies (López de Prado, 2018). Our implementation uses contiguous blocks with an embargo of at least $C \log N$ steps, matching the mixing conditions needed for asymptotic theory.

Mixing and concentration. The theoretical analysis of learning from dependent data is built on strong-mixing (α-, β-) coefficients (Bradley (2005)). Yu (1994) proved the blocking technique that reduces dependent sequences to independent blocks, and McDonald et al. (2015) showed how β-mixing coefficients can be estimated in practice. Matrix concentration for dependent sequences (Banna, Merlevède and Youssef, 2016) is the tool behind the Gram-matrix statements in Chapter 5.

Variance estimation under serial dependence. Heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors (Newey and West (1987); Andrews (1991)) and block resampling (Kunsch (1989); Politis and Romano (1994)) are the classical remedies; our

uncertainty statements use the block bootstrap and HAC bandwidth rules from this literature.

Causal inference for time series. Granger (1969) defined predictive causality, and Sugihara et al. (2012) proposed convergent cross-mapping for coupled dynamical systems; both address *predictability*, not effects of interventions. Runge et al. (2019) combine conditional-independence tests with momentary-conditional-independence to recover lagged causal links, and Runge et al. (2019) review the use of such methods in the Earth sciences, a setting that includes air quality. Marginal structural models (Robins et al. (2000)) handle time-varying treatment under sequential unconfoundedness. Bojinov and Shephard (2019) define causal estimands and randomisation tests for single-unit time-series experiments. Intervention analysis for one treated unit uses synthetic controls (Abadie et al. (2010)) or Bayesian structural time-series counterfactuals (Brodersen et al. (2015)); these need an untreated comparison unit or a pre-period that is stable, which a continuous exposure such as hourly NO_2 does not provide. Lewis and Syrgkanis (2020) treat dynamic treatment effects with g-estimation; Ciganovic et al. (2026) develop DML for time series. All of these works assume the confounders are observed.

2.4 Latent confounders and regime models

Several lines handle unobserved confounding.

- **Representation approaches.** Sequential deconfounders and the time-series deconfounder learn a latent confounder with neural encoders (Bica et al., 2020; Hatt and Feuerriegel, 2021). The cross-sectional ancestors are the substitute-confounder idea of "multiple causes" and its deep-latent-variable relatives (Louizos et al. (2017); Ranganath and Perotte (2018); Veitch et al. (2019); Shi et al. (2019)). Posterior-based adjustment is known to fail without identification (Wang and Blei, 2019; D'Amour, 2019), and Ogburn et al. (2019) sharpen the objection to the substitute-confounder argument. Our results sharpen this caution: *representation quality and downstream reliability are different objects*.
- **Bounds under unobserved confounding.** When identification is lost, effects can still be bounded: Kallus et al. (2019) derive interval estimates of individual-level effects, and Manski (1990) gave the original nonparametric bounds. We do not bound; we characterise the bias that a *specific* generated surrogate leaves behind and report how large a violation would overturn the estimate.
- **Proximal causal inference.** Proxies of an unmeasured confounder identify effects under completeness conditions (Miao, Geng and Tchetgen Tchetgen, 2018; Tchetgen Tchetgen et al., 2020; Mastouri et al., 2021; Cui et al., 2024). The target there is identification; we study what a *plug-in* posterior leaves behind. See Section 2.5.
- **Regime-switching models.** Hamilton (1989) introduced Markov regime switching; Hamilton (1990) gave the likelihood analysis and Chib (1998) Bayesian estimation of multiple change-point models. The hidden-Markov machinery itself is older: the

forward-backward and maximisation steps of Baum and Petrie (1966) and Baum et al. (1970), the EM algorithm of Dempster et al. (1977) and the tutorial of Rabiner (1989). The identifiability and asymptotics of hidden Markov models are classical (Leroux, 1992; Bickel, Ritov and Rydén, 1998; Douc, Moulines and Rydén, 2004; Allman, Matias and Rhodes, 2009); Anandkumar et al. (2014) show that spectral methods identify and estimate latent-variable models from low-order moments, and Gassiat et al. (2016) study inference for finite-state hidden Markov models without parametric emissions. Causal inference for latent Markov models has been studied with the parametric g-formula (Clouth et al., 2025).

- **Stepwise latent-model analysis.** Estimating a latent-class model first and using its posteriors in a second stage is common, and classification error already motivates bias-adjusted corrections (Di Mari, Oberski and Vermunt, 2016). Vermunt (2010) give the three-step approaches with modal and proportional assignment, and Bakk et al. (2013) correct the standard errors of the second step. These adjust for *misclassification of a discrete class*; our setting differs in that the posterior enters a continuous-treatment residual equation, where the error is not a symmetric misclassification. We test the natural two-stage heuristic, not joint likelihood methods.

2.5 Proxies and negative controls

A proxy is a variable that carries information about an unmeasured confounder without being a cause of treatment or outcome; a *negative control* is a proxy known a priori to be unaffected by the treatment (negative-control exposure or outcome). Lipsitch et al. (2010) introduced negative controls as a diagnostic: a "treatment effect" on an outcome that cannot be affected is evidence of confounding. Shi et al. (2020) review the epidemiological use. Formal identification with two proxies (one on each side) is the proximal framework of Miao et al. (2018) and Tchetgen Tchetgen et al. (2020), with doubly negative controls analysed by Shi et al. (2020), panel-data versions by Deaner (2018), and minimax estimation by Kallus et al. (2021). The operational lesson for this project is the diagnostic one: our *placebo* tests (Section 8.3) are negative controls in the sense of Lipsitch et al., 2010, and the threats section (Chapter 11) states which proxies would be needed for point identification, which the data do not provide.

2.6 Omitted-variable bias, amplification and overlap

Our bias laws are omitted-variable identities (Greenland, 1980; Cinelli and Hazlett, 2020; Chernozhukov et al., 2022), and their non-monotonicity is the algebra of *bias amplification*: conditioning on variables that explain the treatment but not the confounder can magnify the residual confounding (Pearl, 2012; Ding, VanderWeele and Robins, 2017). The phenomenon was first noted as a warning against placing instruments in propensity scores (Bhattacharya and Vogt (2007)), analysed with simulations in epidemiology (Myers et al. (2011); Pearl (2011)) and in political science (Middleton et al.

(2016)), and its consequence for matching is discussed by Wooldridge (2016). We claim neither identity as new. What is new is their calibrated, dependent-data and regime-coupled form.

Overlap and irregular identification. Weak overlap is diagnosed in the literature by Gram-matrix or propensity diagnostics (Crump et al., 2009; D'Amour et al., 2021). Its inferential consequences are developed by Khan and Tamer (2010), who show that near-violations of overlap slow convergence and make standard confidence intervals unreliable, by Ma and Wang (2020) and Heiler and Kazak (2021), who give inference that remains valid with many extreme propensity scores, and by Li et al. (2018), who replace inverse weights with overlap weights that are bounded by construction. Stuart (2010) reviews matching and balance checks as design diagnostics. A central finding here is that such conditioning diagnostics can move in the wrong direction when the state is generated.

2.7 Sensitivity analysis

When identification cannot be certified, the standard response is a sensitivity analysis that reports how strong an unmeasured shift would have to be to explain the estimate away (Cinelli and Hazlett, 2020). The line starts with Rosenbaum and Rubin (1983), who parameterised an unobserved binary confounder by its association with treatment and outcome, and continues with Imbens (2003), who gave the partial-R^2 parameterisation in the regression setting. Alternative summaries are the E-value (VanderWeele and Ding (2017)), the coefficient-stability bound of Altonji et al. (2005) as generalised by Oster (2019), and the distributional (selection-model) approach of Tan (2006). Masten and Poirier (2018) study identification when the independence assumption is only partially imposed. For structural causal models, Cinelli et al. (2019) extend sensitivity analysis beyond a single confounder. We adapt this literature to the HMM-regime channel as a one-parameter *robustness value* $b^\star$.

2.8 Air-quality attribution and its causal methodology

This is the applied literature that the data and the question belong to. It splits into five groups.

Exposure and health effects. The classical cohort studies (Dockery et al. (1993); Pope III (2002)) established associations between fine particulate matter and mortality; Dominici et al. (2002) estimated regional and national dose-response relationships with time-series models. Zigler and Dominici (2014) argued that air-pollution epidemiology should move from exposure-response estimation to policy-relevant potential-outcomes questions, and Wu et al. (2020) applied doubly robust and propensity-score methods to long-term $PM_{2.5}$ exposure and mortality in a national cohort. Henneman et al. (2019) link reductions in

coal power-plant emissions to improvements in health outcomes ("accountability" analysis). These studies work with *annual* or *daily* data, health outcomes and rich covariates; they do not address hour-scale pollutant-to-pollutant dynamics.

Burden and trends. Global and national burden estimates (Cohen et al. (2017); Burnett et al. (2018); Lelieveld et al. (2015)) and the India-specific state-level analysis of Balakrishnan et al. (2019) motivate the policy interest. Satellite-derived $PM_{2.5}$ surfaces (Hammer et al. (2020); van Donkelaar et al. (2021)) supply exposure maps where monitors are absent; they are spatial products and do not resolve hourly co-movement at a monitor.

Evaluating interventions. The dominant causal tool in the atmospheric literature is *meteorological normalisation*: a machine-learning model of pollutant concentration on weather and calendar variables is fitted, and the effect of an intervention is read from the change in the weather-normalised series. Grange et al. (2018) developed the random-forest version for Swiss PM_{10}, Grange and Carslaw (2019) formalised its use for detecting interventions, Vu et al. (2019) applied it to the Beijing Clean Air Action, and Shi et al. (2021) to COVID-19 lockdowns in eleven cities. Zhang et al. (2019) attribute the $PM_{2.5}$ decline in China (2013–2017) to emission controls using chemical-transport modelling. For India, regulation and policy have been evaluated by quasi-experimental designs (Greenstone and Hanna (2014)), and the emission inventory of coal-fired plants is documented in Guttikunda and Jawahar (2014). These designs require an *event* (a policy date, a lockdown); the question studied here, how much a continuous change in NO_2 moves $PM_{2.5}$, has none.

Data and monitoring in India. Pant et al. (2019) review particulate monitoring in India and its limits (coverage, calibration, data gaps); Gani et al. (2019) characterise submicron aerosol composition in Delhi and show how strongly secondary aerosol and local combustion vary by season; satellite NO_2 studies (Beirle et al. (2011)) quantify nitrogen-oxide emissions and lifetimes from above. These motivate our data-quality screens (Chapters 6 and 8) and our treatment of NO_2 as one of several co-varying combustion tracers.

Source apportionment and tooling. Receptor-model apportionment (positive matrix factorisation) attributes ambient particulate matter to source categories using chemical speciation; it needs speciated filter data that the CPCB hourly feed does not contain. The `openair` toolkit (Carslaw and Ropkins (2012)) is the standard open implementation of the diagnostics and normalisation used in the literature above. Fann et al. (2012) show how modelled concentration changes are translated into avoided health burden, which is the downstream use of an estimate like ours.

Positioning. To our knowledge, the closest atmospheric methods are the meteorological-normalisation family: they treat weather as the confounder to be removed, with an event as the identifying contrast. The regime view adds a second, *unmeasured* source of co-movement (stagnation episodes that persist for hours to days and are visible only

through meteorology), and asks what happens when that source is estimated rather than observed. We make no claim that NO_2 causes $PM_{2.5}$ in the atmosphere; Chapter 11 states that the estimates are conditional associations whose bias we can characterise but not eliminate.

2.9 Comparison and research gap

Table 2.2. Positioning against the closest lines of work

Line of work	Observed confounders	Dependent data	Latent regime	Exact law for plug-in posterior	Reliability diagnostics
DML (Chernozhukov et al.)	yes	no	no	n/a	n/a
Time-series / dynamic DML	yes	yes	no	n/a	n/a
Deconfounders, sequential encoders	learned	yes	learned	no	accuracy of encoder
Proximal causal inference	proxies	partly	latent	identification only	completeness
Hamilton / HMM regime models	n/a	yes	yes	n/a	likelihood
Three-step latent-class estimation	n/a	no	discrete class	partial (misclassification)	standard-error correction
This work (OR-DML)	yes	**purged block**	**HMM**	**yes**	**residual-state law, $b^\star$, placebos**

Table 2.3. Positioning against the applied air-quality causal toolkit

Method	Needs an event	Handles continuous exposure	Models latent regime	Sensitivity to unmeasured shift	Typical data
Time-series regression (Dominici et al.)	no	yes	no	no	daily counts + pollutants
Meteorological normalisation (Grange, Vu, Shi)	yes	no	no	no	hourly/daily + weather
Synthetic control / CausalImpact	yes	no	no	placebo-in-space/time	panel or reference series
Potential-outcomes cohorts (Wu et al.)	no	yes	no	partial	individual-level health data
This work (OR-DML)	no	**yes**	**HMM**	$b^\star$	hourly pollutant + weather

The gap is therefore not "another estimator", but a quantitative account of what a generated state does to the downstream estimating equation, together with the estimator and reporting rules that account implies. Concretely, the survey leaves four questions open, and each maps to a contribution of this project:

1. *Dependent data with a latent confounder.* DML guarantees need observed confounders, and deconfounder guarantees need identification. What is the exact bias of the plug-in posterior estimator on a dependent series? (Chapter 5.)
2. *Diagnostics that point the wrong way.* Overlap and conditioning diagnostics are validated on observed covariates. Do they remain trustworthy when the covariate is generated? (Chapters 5 and 7.)
3. *Reporting.* Sensitivity analysis exists for a static confounder; what is the one-parameter summary for a regime channel? (Chapters 5 and 8, $b^\star$.)
4. *Applied evidence without an event.* Atmospheric causal studies rely on interventions; what can be said, and with what honesty, for hourly pollutant pairs in Indian cities that have none? (Chapters 8 and 11.)

CHAPTER 3

Problem Formulation

3.1 The structural model

We observe $(Y_t, T_t, X_t, Z_t)_{t=1}^N$ generated by

$$S_t \sim \text{Markov}(\Pi, \pi_0), \qquad Z_t = \mu_Z(S_t) + \xi_t, \tag{3.1}$$

$$T_t = m(X_t, S_t) + V_t, \qquad Y_t = \theta(S_t)\, T_t + g(X_t, S_t) + U_t, \tag{3.2}$$

with $S_t \in \{1, \dots, K\}$ latent, $\mathbb{E}[V_t \mid X_t, S_t] = 0$, $\mathbb{E}[U_t \mid X_t, S_t, T_t] = 0$, and $\theta(S_t) = \sum_k \theta_k^* 1(S_t = k)$. The targets are the regime effects θ_k^* and the stationary average $\theta_{\text{ATE}}^* = \sum_k \pi_k^* \theta_k^*$. In the application T is NO_2, Y is $PM_{2.5}$, X collects lagged pollutants and weather, and Z is meteorology.

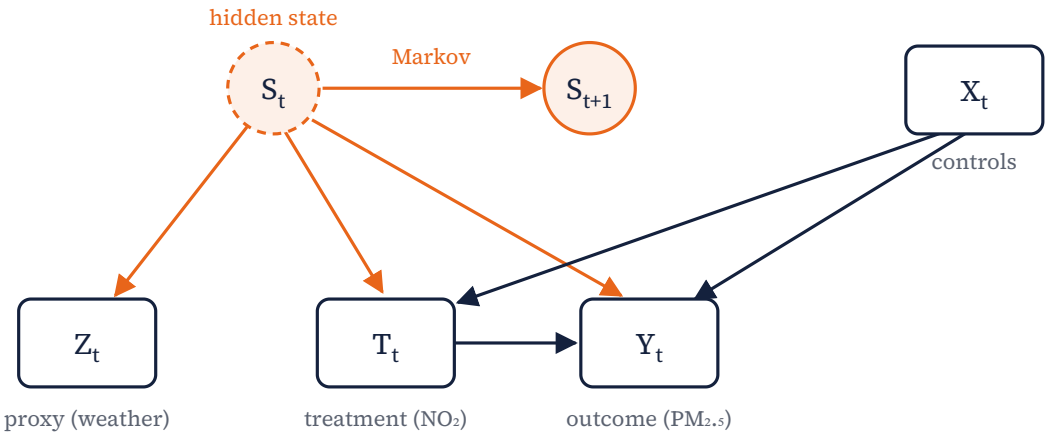


Figure 3.1. Causal structure. The latent state S_t evolves as a Markov chain, is seen only through the proxy Z_t, and moves both treatment T_t and outcome Y_t, so it is a confounder that conditioning on X_t alone cannot remove.

Regime-conditional unconfoundedness, overlap and positivity, bounded variables, geometric α-mixing, and the usual $o_P(N^{-1/4})$ nuisance rates are stated formally in Annexure G (A1 to A4).

3.2 Why standard DML fails

Standard DML that residualizes on X_t alone cannot see S_t. The following result states exactly what it estimates.

> **Theorem 1 (Omitted regime bias)**
> Standard DML that residualizes on X_t alone satisfies $\hat{\theta}_{\text{naive}} \to_p \sum_k \omega_k \theta_k^* + \mathcal{B}_{\text{het}} + \mathcal{B}_{\text{conf}}$, with weights $\omega_k = \mathbb{E}[1(S_t{=}k)\tilde{T}_t^2]/\mathbb{E}\tilde{T}_t^2$ and
>
> $$\mathcal{B}_{\text{conf}} = \frac{\sum_k \mathbb{E}[\pi_k(X)\,\Delta m_k(X)\,\Delta g_k(X)]}{\mathbb{E}[\tilde{T}_t^2]},$$
>
> where $\Delta m_k, \Delta g_k$ are the regime deviations of the treatment mean and the outcome baseline.

The mechanism is intuitive. The numerator, the covariance between the regime's effect on the treatment and on the outcome, does not depend on how well X explains T. The denominator, the residual treatment variance $\mathbb{E}\tilde{T}_t^2$, *shrinks* as X explains more of T. Residualizing on more observed controls can therefore **amplify** omitted-regime bias and even flip its sign (Pearl, 2012). In the synthetic frontier of Chapter 7, standard DML has bias 8.44 on a true ATE of 1.53 at every level of proxy quality.

3.3 Non-identification without state information

The opposite extreme is a proxy with no information about the state.

> **Theorem 2 (Non-identification without state information)**
> If $P(Z_t \mid S_t{=}j) = P(Z_t \mid S_t{=}k)$ for all j, k and transitions are memoryless, there exist two models satisfying all assumptions, with identical laws of $(Y, T, X, Z)_{1:N}$ for every N and distinct, non-permuted effect vectors.

The construction used in the final proof is an assumption-consistent randomized-treatment design: independent randomized treatment, independent memoryless states, and uninformative proxies. Its two binary-outcome models have identical complete observable laws and different regime-effect vectors, while their *average* effect is shared and identified (Chapter 10 describes the correction that produced this form). The exact probability tables are enumerated by `src/synthesis/nonidentification_check.py` and verified by `verify_submission.py`.

Between the two extremes lies the regime that matters in practice: proxies that are *informative but imperfect*. The rest of the report studies that regime.

3.4 What "reliable" must mean

The reliability rule in current practice is built from two quantities that are both easy to compute.

- the **posterior error** $\varepsilon_\gamma = \mathbb{E}\|\gamma_t - H_t\|_1$ between the posterior and the one-hot true state, whose observable proxy is the posterior entropy;
- the **conditioning** of the score Jacobian, $\lambda_{\min}(\hat{J})$.

Both appear in error bounds of the form $C\varepsilon_\gamma/\lambda_{\min}(J)$ (Theorem 3 in Chapter 5), and in a controlled synthetic design $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$ ranks causal error almost perfectly (Spearman 0.995 over 4,900 runs). The research question is whether that success survives when posteriors are learned and the state is generated, and, if it does not, what replaces it.

CHAPTER 4

Overlap-Aware Regime DML: Method and Implementation

4.1 Design principle

The estimator is built from one observation: the posterior enters the downstream estimating equation **twice**. It acts as an *adjustment feature* inside the nuisance regressions, where it supplies fine-grained information about the state, and as a *weight* in the score that defines the regime effects, where it selects a coarse target. The failure analysis of Chapter 5 shows that these two roles must be kept apart: using the posterior as a weight on the nuisance fit leaks regime contrasts into the score, while using it as a feature removes them up to what the posterior cannot explain.

4.2 The estimator

Step 1: latent-state inference. A Gaussian HMM is fitted to the proxies Z by EM with multiple restarts. Posteriors are either smoothed, $\gamma_{tk} = P(S_t{=}k \mid Z_{1:N})$, or causally filtered, $P(S_t{=}k \mid Z_{1:t})$.

Step 2: purged block cross-fitting. The series is split into B contiguous blocks. For each block, nuisances are fitted on the other blocks after dropping an embargo of τ^* steps on each side of the evaluation block (Figure 4.1). The embargo is $\tau^* = \max\{h : \max_j |\hat{\rho}_j(h)| \geq 1.96/\sqrt{N}\} + 1$, and never less than $C \log N$, a floor required by the mixing argument.

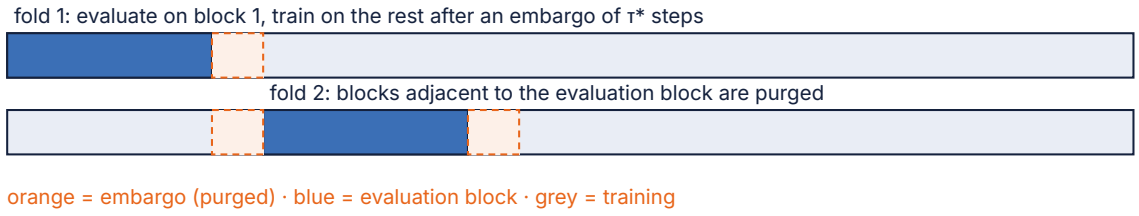


Figure 4.1. Purged block cross-fitting. Training observations within the embargo τ^* of the evaluation block are dropped so that temporal dependence cannot leak between folds.

Step 3: posterior-adjusted nuisances. One pair of nuisances is cross-fitted on the posterior-augmented features $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$: $\hat{m}(F) \approx \mathbb{E}[T \mid F]$ and $\hat{\mu}(F) \approx \mathbb{E}[Y \mid F]$. With $\tilde{T}_t = T_t - \hat{m}(F_t)$ and $\tilde{Y}_t = Y_t - \hat{\mu}(F_t)$ the coupled system is

$$\hat{J}_{jk} = \frac{1}{N}\sum_t \gamma_{tj}\gamma_{tk}\,\tilde{T}_t^2, \qquad \hat{S}_k = \frac{1}{N}\sum_t \gamma_{tk}\,\tilde{T}_t\tilde{Y}_t, \tag{4.1}$$

$$\hat{\boldsymbol{\theta}}_\lambda = (\hat{J} + \lambda I)^{-1}\hat{S}. \tag{4.2}$$

Step 4: spectral regularization. The penalty $\lambda \geq 0$ stabilizes the inversion when $\lambda_{\min}(\hat{J})$ is small. On a documented path it reduces variance sharply but introduces bias, so λ is a tuning knob, not free lunch (Section 7.6).

Step 5: inference. A HAC sandwich variance uses the observation-level moments actually solved, including off-diagonal terms. A residual bootstrap uses the same moment matrices as the point solve.

> **Algorithm 1: OR-DML with a reliability report**
> **Input:** $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, number of folds B, penalty λ.
>
> 1. Fit a Gaussian HMM to Z (multi-restart EM); compute posteriors γ_t (smoothing or causal filter).
> 2. Choose embargo $\tau^* = \max\{h : \max_j |\hat{\rho}_j(h)| \geq 1.96/\sqrt{N}\} + 1$, at least $C \log N$.
> 3. Cross-fit $\hat{m}, \hat{\mu}$ on $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$ over purged contiguous blocks.
> 4. Form $\hat{J}, \hat{S}$ by (4.1); return $\hat{\theta}_\lambda = (\hat{J} + \lambda I)^{-1}\hat{S}$ with a HAC sandwich.
> 5. **Report:** hard-FE and posterior-feature baselines; $\hat{v}$, $\hat{a}$, $\widehat{\mathrm{Var}}(\tilde{T})$ and the robustness value $b^\star$; raw and standardized $\lambda_{\min}$; lead placebos aligned to the outcome's information set.

Complexity is $\mathcal{O}(B\,C_{\text{nuis}} + NK^2 + K^3)$, where C_{nuis} is the cost of one nuisance fit. A scalability experiment checks $K \leq 5$ regimes (Section 7.7).

4.3 Baselines

Two simpler estimators on the same inputs serve as the main baselines, alongside standard DML, block DML and DML with Z as extra controls.

- **Hard regime fixed effects:** one-hot $\arg\max$ weights, the practitioner's default after HMM inference.
- **Posterior-feature DML:** a single-index DML whose nuisances are on F_t, with no regime-coupled score.
- **OR-DML with weighted nuisances:** the design of earlier project drafts, in which regime-k nuisances are fitted on X with weights $\gamma_{\cdot k}$. It is kept as an *ablation*; its Gram part is not orthogonal for soft weights (Section 5.1).

4.4 Software architecture

The estimator is implemented in `src/or_dml.py` (648 lines) in scikit-learn style. The package splits into three decoupled parts, shown with the classes that implement them.

Table 4.1. Software components of the estimator

Component	Class / function	Role
Latent-state inference	`LatentRegimeHMM`	multi-restart Gaussian-HMM EM; smoothed and filtered posteriors; BIC
Temporal validity	`estimate_optimal_embargo`, `PurgedBlockKFold`	embargo choice with a $C \log N$ floor; purged contiguous folds
Coupled estimator	`OverlapAwareRegimeDML`	nuisance cross-fitting, $\hat{J}$, $\hat{S}$, spectral solve, HAC and bootstrap uncertainty, dynamic IRF
Regularization	`AdaptiveSpectralOptimizer`	data-driven λ with an oracle risk bound

The *posterior* nuisance mode, which implements the paper's design, builds the augmented feature matrix and fits one pair of nuisances per fold (full listings in Annexure C):

```
G = weights[:, 1:]                    # posterior probabilities, reference regime dropped
F = np.column_stack([X, G] + [G[:, [k]] * X for k in range(G.shape[1])])
for train_idx, test_idx in splitter.split(N):      # purged blocks
    m_y = clone(self.nuisance_model).fit(F[train_idx], Y[train_idx])
    m_t = clone(self.nuisance_model).fit(F[train_idx], T[train_idx])
    ry = Y[test_idx] - m_y.predict(F[test_idx])    # out-of-fold residuals
    rt = T[test_idx] - m_t.predict(F[test_idx])
```

and the coupled solve forms the Gram matrix and applies the spectral penalty:

```
for j in range(K):
    S_vec[j] = np.mean(w[:, j] * tilde_T[j] * tilde_Y[j])
    for k in range(K):
        J_mat[j, k] = np.mean(w[:, j] * w[:, k] * tilde_T[j] * tilde_T[k])
theta_vec = np.linalg.inv(J_mat + eff_lambda * np.eye(K)) @ S_vec
```

4.5 Verification of the implementation

Regression tests in `src/test_inference_regression.py` cover the four score/bread mode combinations, score-mean invariance of the HAC, the occupation cross-covariance, bootstrap moment consistency and the uncapped logarithmic embargo floor. Each defect that these tests guard was first reproduced against the earlier implementation (Chapter 10). `verify_science.py` and `verify_submission.py` provide higher-level checks (Chapter 9).

CHAPTER 5

Theory: How State Uncertainty Propagates

Let $H_t = e_{S_t}$ be the one-hot true state, let $\varepsilon_\gamma = \mathbb{E}\|\gamma_t - H_t\|_1$ be the posterior error, and let J, S be the population versions of the coupled system. Define $\theta_\gamma = J^{-1}S$ and $b_\gamma = S - J\theta^*$, so that $\theta_\gamma - \theta^* = J^{-1}b_\gamma$ exactly.

5.1 Exact orthogonality

> **Lemma 1 (Exact orthogonality)**
> Let $\psi_k(\theta; m, \mu) = \gamma_k(T - m)\{Y - \mu - \sum_j \gamma_j\theta_j(T - m)\}$. At $m = \mathbb{E}[T \mid F]$ and $\mu = \mathbb{E}[Y \mid F]$, the Gateaux derivative of $\mathbb{E}\psi_k$ in every direction (δ_m, δ_μ) of square-integrable functions of F vanishes, for every θ and every posterior γ; so does that of $\mathbb{E}[\gamma_j\gamma_k(T - m)^2]$.

Proof. Each derivative is an expectation of a product of an F-measurable factor (the weights γ and the perturbation δ) and either $T - m$ or $Y - \mu - \sum_j \gamma_j\theta_j(T - m)$, whose conditional mean given F is zero. □

The natural alternative, used in earlier drafts, fits regime-k nuisances on X with weights $\gamma_{\cdot k}$. Its Gram part is **not** orthogonal for soft weights: under a nuisance perturbation its score derivative is 85% of the Jacobian, against 0.4% for the posterior-adjusted design of Lemma 1. This is the formal reason OR-DML conditions on the posterior rather than weighting by it.

5.2 Graceful degradation

> **Theorem 3 (Graceful degradation)**
> Suppose the nuisance targets are within $C_\eta\varepsilon_\gamma$ of the oracle regime nuisances in L^1, and $\lambda_{\min}(J) > 0$. Then $\|\theta_\gamma - \theta^*\|_2 \le C\varepsilon_\gamma/\lambda_{\min}(J)$ and
>
> $$\|\hat{\theta}_\gamma - \theta^*\|_2 \le \frac{C\varepsilon_\gamma}{\lambda_{\min}(J)} + \mathcal{O}_P\Big(\frac{N^{-1/2} + \rho_N}{\lambda_{\min}(J)}\Big),$$
>
> with $\rho_N = r_N^2$ for posterior-adjusted nuisances and $\rho_N = r_N$ for posterior-weighted ones, where r_N is the nuisance rate.

The bound separates the two errors that the problem involves: **state recovery** (ε_γ) and **nuisance estimation** (ρ_N). Posterior adjustment improves the nuisance term from first to second order. In a controlled design the ratio $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$ ranks causal error almost

perfectly (Spearman 0.995 over 4,900 runs, against 0.626 for ε_γ alone and 0.521 for $1/\lambda_{\min}$). The next results explain why that success is fragile: $\lambda_{\min}(J)$ is itself a function of the state error.

5.3 The coupled score under state confusion

Consider two regimes with stationary weights π_s, treatment means m_s, outcome means $\mu_s = \theta_s m_s + g_s$ and innovation scales σ_s. Take regime-wise nuisances and a proxy that places weight $1-\alpha$ on the true regime and $\alpha = \varepsilon_\gamma/2$ on the other.

Proposition 1 (Exact law for regime-weighted scores)
With $\gamma_j(s) = 1-\alpha$ if $j = s$ and α otherwise,

$$J_{jk} = \sum_s \pi_s \gamma_j(s)\gamma_k(s)\big[(m_s - m_j)(m_s - m_k) + \sigma_s^2\big],$$

$$S_j = \sum_s \pi_s \gamma_j(s)\big[(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2\big].$$

In the symmetric case ($\pi_s = \frac{1}{2}$, $\sigma_s = \sigma$, $\Delta = m_1 - m_0$, $G = \mu_1 - \mu_0$), J has eigenvectors $(1, \pm 1)$ with eigenvalues $\lambda_+ = \frac{1}{2}\sigma^2 + \frac{1}{2}\alpha^2\Delta^2$ and $\lambda_- = \frac{1}{2}\sigma^2(1-2\alpha)^2 + \frac{1}{2}\alpha^2\Delta^2$.

Corollary 1 (Phantom resolution and the error hump)
In the symmetric case: **(a)** $\lambda_{\min}(J) = \lambda_-$ *exceeds* its oracle value $\sigma^2/2$ whenever $\varepsilon_\gamma > 8\sigma^2/(\Delta^2 + 4\sigma^2)$, so a confused state can look *better* conditioned than the true one; **(b)** when $|\Delta G| \gg \sigma^2|\theta_0 - \theta_1|$ the error is $\approx \sqrt{2}\,\frac{\varepsilon_\gamma}{2}|\Delta G|/(\sigma^2 + \varepsilon_\gamma^2\Delta^2/4)$, maximized at $\varepsilon^\star = 2\sigma/|\Delta|$ and *decreasing* beyond it; **(c)** the dominant displacement lies on the λ_+ eigenvector, so the worst-direction bound of Theorem 3 is loose by λ_+/λ_-.

Proposition 1 is exact and involves no sampling. Applied to the asymmetric factorial design it reproduces every cell's median error to within 0.98–1.02 times the observed value and locates the error peaks at $\varepsilon_\gamma \in \{0.05, 0.10, 0.20\}$ for $\sigma_1 \in \{0.08, 0.15, 0.30\}$, exactly where they are observed (Figure 5.1). Mixing regimes leaks between-regime treatment contrasts into the residualized treatment, which inflates the Gram matrix. The factorial's near-perfect rank correlation is therefore an *algebraic identity of the design*, not a statistical regularity: replacing $\lambda_{\min}(\hat{J})$ by the design value σ_1^2 lowers the cell-level correlation from 0.997 to 0.958.

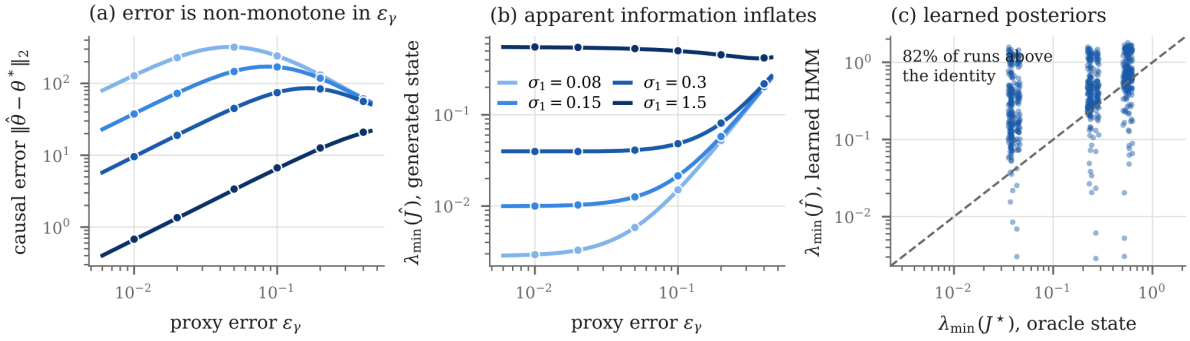


Figure 5.1. Phantom causal resolution. (a) Coupled regime-score error against proxy error ε_γ for four treatment-innovation scales σ_1 (dots: median of 100 runs of the 7×7 factorial; lines: the exact law of Proposition 1, no fitted parameters). Error is non-monotone, peaking near $\varepsilon^\star \approx 2\sigma/|\Delta_m|$. (b) The smallest eigenvalue of the generated-state Gram matrix: unless σ_1 is large it *grows* with proxy error (70-fold at $\sigma_1 = 0.08$). (c) With posteriors learned by EM (720 runs) $\lambda_{\min}(\hat{J})$ exceeds the oracle-state value in 82% of runs.

5.4 Posterior-adjusted designs and the residual-state bias law

Posterior-adjusted nuisances remove the contrasts that weighted nuisances leak, up to what the posterior cannot explain. Let $v_t = \gamma_t(1-\gamma_t)$ be the posterior variance of the regime indicator, and let $T_t = m(X_t) + aS_t + V_t$ and $Y_t = \theta_{S_t} T_t + g(X_t) + bS_t + U_t$, with treatment regime shift a and outcome regime shift b.

> **Theorem 4 (Residual-state bias law)**
> Let $K = 2$ and suppose the posterior is calibrated given F, V_t is independent of (S, Z, X) with variance σ^2, nuisances are cross-fitted at rates with $r_N^2 = o(N^{-1/2})$, and the posterior forgets distant proxies. With score weights $w_t = (1-\gamma_t, \gamma_t)$ (OR-DML) or $w_t \equiv 1$ (posterior-feature DML),
>
> $$\hat{\theta} \to_p \bar{\theta} + J^{-1} b_\gamma, \qquad J = \mathbb{E}\big[ww^\top(a^2 v + \sigma^2)\big], \quad b_\gamma = a\,\mathbb{E}[\,w\,c\,v\,],$$
>
> with $c_t = b + (\theta_1 - \theta_0)\{m(X_t) + a(1-\gamma_t)\}$ and $\bar{\theta} = \theta^*$ for OR-DML. Hence: **(a)** with no treatment regime shift ($a = 0$) both are consistent for any calibrated posterior, whatever b and the effect heterogeneity; **(b)** for homogeneous effects the pooled bias is $abv/(a^2 v + \sigma^2)$, hump-shaped in a with peak at $|a| = \sigma/\sqrt{v}$ and $\sup|B| = |b|\sqrt{v}/(2\sigma)$, increasing in v, and a soft posterior feature is never worse than its hard label; **(c)** J grows with $a^2 v$, so phantom resolution is not specific to the weighted design.

The law is exact up to sampling error. Over 9 configurations with $N = 10^6$ the estimator and the law agree to within 0.008, and with $a = 0$ the bias is 0.005. With learned HMM posteriors, the plug-in form of the law matches measured bias across 72 cells with correlation 0.996 and median relative error 5.2% (Figure 5.2).

Both laws have the same anatomy: a numerator proportional to *state error times the product of the treatment and outcome regime shifts*, divided by a *residual treatment variance* that the same treatment shift inflates. This is why $\mathrm{Var}(\tilde{T})$, the scalar analogue of $\lambda_{\min}(\hat{J})$, cannot be read as a monotone certificate.

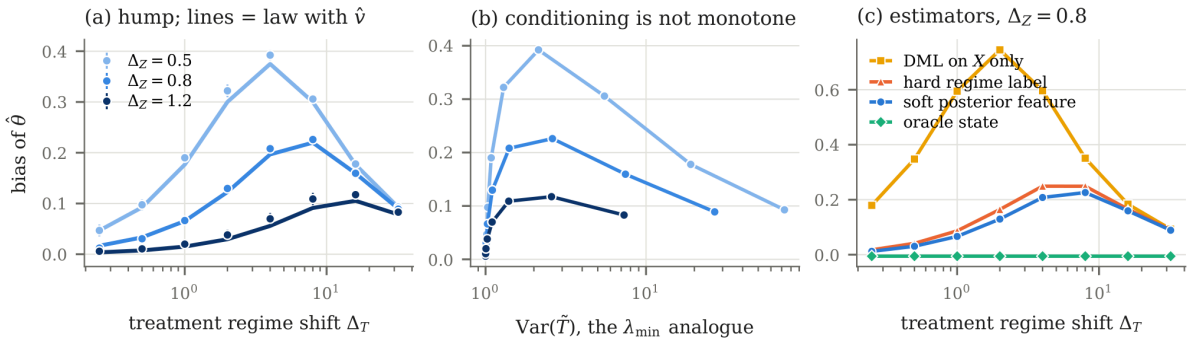


Figure 5.2. The residual-state bias law with learned HMM posteriors. (a) Measured bias is hump-shaped in the treatment regime shift and matches the law evaluated with the observable plug-in $\hat{v}$ (lines). (b) The same bias against $\mathrm{Var}(\tilde{T})$: conditioning improves monotonically while bias first rises and then falls. (c) Soft posterior features beat hard labels; DML on X alone is far worse; the oracle state is unbiased.

5.5 Observability, sensitivity and inference

For a calibrated posterior, $\hat{v} = N^{-1}\sum_t \hat{\gamma}_t(1-\hat{\gamma}_t)$ satisfies $|\mathbb{E}\hat{v} - v| \le \sqrt{CE} + CE$ in the L^2 calibration error CE, and $\mathrm{Var}(\tilde{T})$ is estimable. If the calibrated regime-mean model identifies the treatment shift a, only the outcome shift b remains unidentified, so the **robustness value**

$$b^\star = \frac{|\hat{\theta}|\,\widehat{\mathrm{Var}}(\tilde{T})}{\|\hat{\Sigma}\hat{a}\|}$$

(the smallest outcome shift that explains the estimate away) gives a one-parameter sensitivity analysis. Under misspecification $\hat{v}$ may understate or overstate the residual state variance, so sensitivity values require calibration assumptions.

> **Theorem 5 (Inference around the proxy target)**
> For posterior-adjusted nuisances, under Theorem 3 and boundedness, augmented-process β-mixing, nuisance-rate and differentiability conditions, with $\tau^* \ge C\log N$ and a separated HMM calibration window, $\sqrt{N}(\hat{\theta}_\lambda - \theta_\lambda) \to \mathcal{N}(0, \Sigma_\lambda^{\mathrm{tot}})$, including representation uncertainty. Here $\theta_\lambda - \theta^* = (J+\lambda I)^{-1}(b_\gamma - \lambda\theta^*)$ and $\|b_\gamma\| \le C\varepsilon_\gamma$. Structural coverage requires displacement $o(N^{-1/2})$.

When proxy displacement is not negligible, structural Wald tests need not have nominal size. The report therefore pairs estimates with sensitivity analysis rather than an oracle bias correction. Implemented HAC intervals condition on the fitted HMM; **full-sample smoothing lies outside this theorem**, a limit recorded explicitly in the manuscript.

5.6 Summary of what the theory says

Table 5.1. Summary of the theoretical results

Result	Statement	Practical consequence
Theorem 1	Standard DML is biased by omitted regimes; more controls can amplify	do not rely on standard or block DML
Theorem 2	No state information gives non-identified regime effects	proxies must carry information
Lemma 1	Posterior-adjusted coupled score is exactly Neyman-orthogonal	condition nuisances on the posterior
Theorem 3	Error $\leq C\varepsilon_\gamma/\lambda_{\min}$ plus nuisance terms	state and nuisance errors separate
Proposition 1, Cor. 1	Exact error hump and phantom resolution	$\lambda_{\min}$ is not a monotone certificate
Theorem 4	Residual-state bias law	report $\hat{v}$, regime shifts, $b^\star$
Theorem 5	Asymptotic normality around the proxy target	intervals conditional on fitted HMM only

CHAPTER 6

Data and Experimental Design

6.1 Data sources

The empirical work uses two open sources.

- **OpenAQ / CPCB.** Continuous ambient air-quality records from monitoring stations of India's Central Pollution Control Board, accessed through the OpenAQ platform: $PM_{2.5}$, PM_{10}, NO, NO_2, NO_x, CO, SO_2, O_3 and station-reported meteorology.
- **Open-Meteo.** Hourly ECMWF ERA5 reanalysis: temperature, humidity, precipitation, wind speed and direction, pressure.

The raw hourly files are stored in `data/raw_hourly/`. Pollution files exist for four cities, and weather files for those four plus three further cities that the earlier applied study used (Chennai, Hyderabad, Ahmedabad).

Table 6.1. Raw pollution records per city (all parameters, half-hourly UTC stamps)

City	Rows	First reading (UTC)	Last reading (UTC)
Delhi	123,323	2025-02-18	2026-06-16
Mumbai	61,628	2025-02-18	2026-06-16
Bengaluru	94,713	2025-02-18	2026-06-16
Kolkata	22,847	2025-02-18	2025-06-23

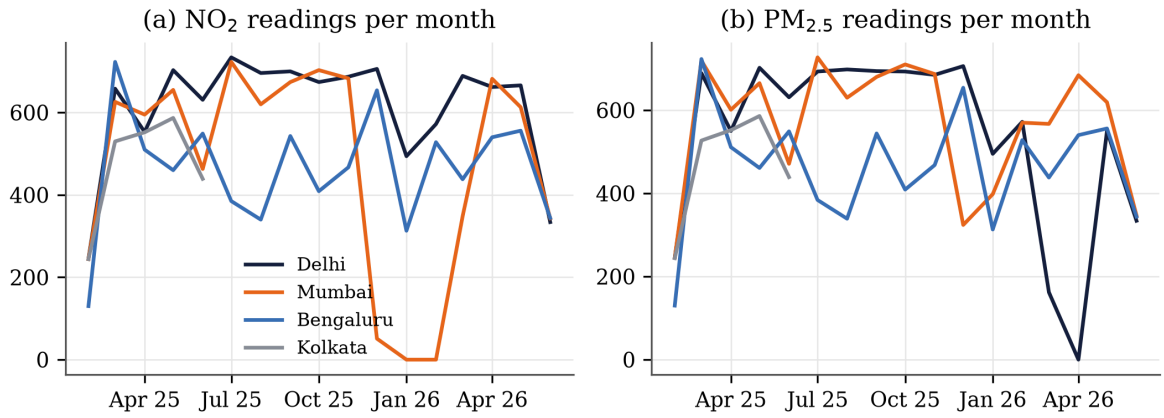


Figure 6.1. Data coverage. Monthly count of NO_2 and $PM_{2.5}$ readings by city. Kolkata's record ends in June 2025; sparse coverage (and, in Mumbai, long constant runs, Section 8.2) are why the four cities behave so differently in Chapter 8.

6.2 Variable construction

The treatment T is NO_2 ($\mu g/m^3$), the outcome Y is $PM_{2.5}$ ($\mu g/m^3$), the regime proxies Z are meteorological series (temperature, humidity, wind speed and direction, pressure), and the controls X are lagged pollutants and lagged weather. NO_2 is a combustion marker whose emission is dominated by traffic and industry, which is why it serves as an emission-intensity proxy for treatment; the associated limitation, that NO_2 is itself chemically related to $PM_{2.5}$ formation, is stated in Section 8.5.

6.3 The time-grid audit

A data-quality audit found that the legacy panel had rounded half-hour timestamps with ties-to-even, collapsing the series onto a **two-hour grid**. What had been labelled "hourly" lags were therefore two hours, and every horizon, lag length and embargo built on them was mis-scaled. The effect was not cosmetic: the Delhi single-index posterior-feature estimate moved from 0.58 to 0.37 once the grid was corrected. The corrected pipeline (`src/data_pipeline_clean.py`) aggregates properly to an hourly grid, and all real-data numbers in the report are recomputed on it.

> The lesson recorded in the project: *a correct estimator on a mis-specified grid produces confident wrong numbers.* The audit is the reason the manuscript contains a provenance appendix.

6.4 Experimental designs

The evidence is deliberately layered, from fully known truth to fully observational.

Table 6.2. Experimental designs and what each can establish

Design	Truth known?	What it tests	Size
Factorial (7×7)	yes	closed-form law; rank correlation of failure scores	4,900 runs, 42 cells
Learned-posterior grid	yes	removes the circularity of synthetic proxies; EM posteriors	720 runs, 36 cells, $N = 1{,}200$
Synthetic frontier	yes	observability (Δ_Z) $\times$ persistence	900 datasets
Residual-state law checks	yes	Theorem 4 against estimator, $N = 10^6$	9 configurations; 72 learned-posterior cells
Real-data-calibrated semi-synthetic	effects known; everything else real	four cities, real gaps, meteorology, NO_2 series	30 datasets per city
Representation zoo	yes	do better encoders give better causal reliability?	15 shifted worlds, 4 encoders
Observational stress test	**no**	audited associations, placebos, robustness values	four cities

Factorial design. A 7×7 grid over proxy error ε_γ and treatment-innovation scale σ_1 with 100 runs per cell controls the posterior error directly, so the closed form of Proposition 1 can be checked cell by cell.

Learned posteriors. To remove the circularity of synthetic proxies, posteriors are learned with a Gaussian HMM (EM, three restarts, smoothing) from noisy proxies across proxy separation Δ_Z, innovation scale σ_1 and treatment shift Δ_T. All diagnostics use the *realized* generated-state Gram matrix.

Real-data-calibrated semi-synthetic benchmark. The benchmark keeps everything real except the causal effects: the meteorological proxies and controls, the observed NO_2 series, the hourly missingness pattern, and regime persistence and overlap. True state paths are drawn by forward-filtering backward-sampling from an HMM calibrated per city (stay probabilities 0.90–0.96), so truth is known while the structure is realistic. It replaces an earlier benchmark in which the true state was a deterministic function of the proxies.

6.5 Estimators compared and metrics

Estimators: oracle-state DML (infeasible lower bound), standard DML, block DML, DML with Z controls, posterior-feature DML, hard regime fixed effects, OR-DML with weighted nuisances (ablation) and posterior-adjusted OR-DML. Metrics are the mean absolute ATE bias, the median regime-effect error $\|\hat{\theta} - \theta^*\|_2$, RMSE, 95% interval coverage, posterior error ε_γ, and the raw and standardized $\lambda_{\min}$.

CHAPTER 7

Results: Synthetic and Semi-Synthetic Evidence

7.1 The closed form reproduces the factorial experiment

Proposition 1 predicts the coupled-score error without fitting any parameter. Across the 42 cells of the factorial design (4,900 runs) the ratio of observed to predicted median error has median 0.999 (interquartile range 0.993–1.008, full range 0.98–1.02). The error peaks sit where the law places them (Figure 5.1a), and the generated-state $\lambda_{\min}$ inflates up to 70-fold (Figure 5.1b). The failure score $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$ achieves Spearman 0.995 with causal error, but, as Section 5.3 showed, partly *because* $\lambda_{\min}$ inflates.

7.2 Learned posteriors break the scalar ratio

When posteriors are learned by EM from noisy proxies (720 runs, 36 cells), the picture changes.

Table 7.1. Spearman correlation between failure scores and causal error. "Synthetic proxy" is the factorial design; "learned" uses EM posteriors; "within Δ_T" averages correlations at a fixed treatment shift.

	synthetic proxy	learned HMM posteriors		
Failure score	**runs**	**runs**	**within Δ_T**	**cells**
ε_γ (posterior error)	0.626	0.187	0.083	0.222
$1/\lambda_{\min}(\hat{J})$	0.521	0.465	0.483	0.509
$\varepsilon_\gamma/\lambda_{\min}(\hat{J})$	0.995	0.517	0.388	0.564
$\bar{H}/\lambda_{\min}(\hat{J})$ (observable)	–	0.581	0.442	0.611
$\varepsilon_\gamma/\lambda_{\min}(J^\star)$ (oracle geometry)	–	0.750	0.727	0.783
$\bar{H}\lvert\hat{\Delta}_T\hat{\Delta}_Y\rvert/\lambda_{\min}(\hat{J})$ (observable)	–	0.664	0.730	0.675
$\varepsilon_\gamma\lvert\hat{\Delta}_T\hat{\Delta}_Y\rvert/\lambda_{\min}(\hat{J})$	–	0.695	0.720	0.863

- The ratio $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$ falls from **0.995 to 0.517**.
- Restoring the *oracle* geometry recovers 0.750, so the damage is in the generated-state Gram matrix.
- Restoring instead the numerator scale that the laws predict, using the plug-in regime shifts $\hat{\Delta}_T, \hat{\Delta}_Y$, gives 0.695 (0.863 across cells), and 0.664 with entropy in place of ε_γ, which is *fully observable*.
- With the same posteriors, hard fixed effects beat the posterior-weighted design in 100% of cells, while posterior-adjusted OR-DML beats hard fixed effects in 72% of cells (median error ratio 0.97; 0.69 against the weighted design). All its losses occur at the smallest treatment innovation, where the a^2v term dominates J.

> The actionable replacement for the scalar rule is therefore a *product*: representation uncertainty times the regime shifts it multiplies, divided by conditioning (Chapter 11 develops this "dual-factor" view).

7.3 Observability and persistence frontier

Table 7.2 (left) varies proxy separation Δ_Z in the data-generating process of Theorem 1; a persistence sweep adds regime-0 stay probabilities 0.70 and 0.97 (900 datasets in all). Standard and blocked DML are stuck at the omitted-regime bias (8.44 and 8.47 against a true ATE of 1.53). All proxy-based estimators fail at $\Delta_Z \leq 1$ and approach the oracle at $\Delta_Z \geq 2$.

Table 7.2. Posterior-adjusted OR-DML is the most accurate regime-aware estimator, but no estimator removes residual-state bias. Left: synthetic frontier with learned HMM posteriors (100 datasets per Δ_Z, $N = 1{,}200$, persistence 0.88, true ATE 1.53). Right: real-data-calibrated semi-synthetic benchmark on the corrected hourly grid (30 datasets per city). Bold: best non-oracle within each block.

	synthetic, proxy separation Δ_Z			real-data-calibrated, hourly			
Estimator	**1**	**2**	**4**	**Del**	**Mum**	**Ben**	**Kol**
ATE: mean absolute bias							
Oracle-state	0.03	0.03	0.03	0.03	0.07	0.12	0.14
Standard DML	8.43	8.44	8.44	1.94	3.00	2.97	3.23
Block DML	8.47	8.47	8.48	2.16	3.09	3.20	3.30
DML + Z controls	5.43	2.63	0.86	1.22	3.03	2.60	3.06
Posterior-feature DML	2.16	0.14	0.04	0.55	**1.03**	**1.51**	2.17
Hard regime FE	2.57	0.14	**0.03**	0.55	1.86	1.97	2.30
OR-DML, weighted nuisances	4.24	0.15	**0.03**	0.88	3.10	3.22	3.81
OR-DML (posterior-adjusted)	**2.13**	**0.13**	**0.03**	**0.42**	1.65	1.69	**2.05**
Regime effects: median $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|_2$							
Hard regime FE	3.65	0.08	**0.06**	0.77	2.84	2.83	3.21
OR-DML, weighted nuisances	6.11	0.10	0.06	1.21	4.59	4.56	5.33
OR-DML (posterior-adjusted)	**2.95**	**0.07**	0.06	**0.59**	**2.56**	**2.40**	**2.88**
posterior error ε_γ	0.062	$<$0.001	$<$0.001	0.087	0.097	0.088	0.083

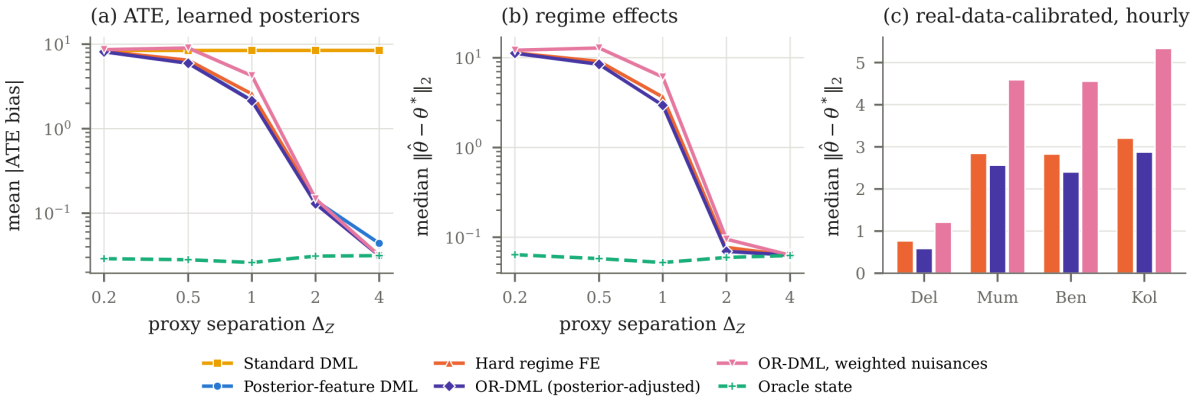


Figure 7.1. Conditioning nuisances on the posterior, rather than weighting them by it, fixes the coupled estimator. (a) Mean absolute ATE bias and (b) median regime-effect error along the observability frontier with learned posteriors. (c) Regime-effect error in the real-data-calibrated benchmark: the same ordering in every city.

Among the 7 cells with $\Delta_Z \geq 1$, OR-DML is best or tied for the ATE in 7 of 7 and for regime effects in 7 of 7. At $\Delta_Z = 2$ its ATE bias is 0.129 (hard fixed effects 0.141, weighted design 0.147) with coverage 91%; at $\Delta_Z = 1$ its regime-effect error is 2.954 against 3.652 for hard fixed effects and 6.107 for the weighted design. Persistence helps exactly through the posterior: at $\Delta_Z = 1$, raising the stay probability from 0.70 to 0.97 lowers ε_γ from 0.090 to 0.026 and OR-DML's bias from 2.725 to 0.960. Under a continuous latent mixture that no two-state model describes, OR-DML beats hard fixed effects in 100% of 100 replications, while the weighted design beats them in 0%.

7.4 Real-data-calibrated semi-synthetic benchmark

The right half of Table 7.2 keeps everything real but the causal effects. With posteriors that are exactly calibrated and 95–96% accurate, every proxy-based estimator retains large bias and zero interval coverage. For regime effects OR-DML is the most accurate regime-aware estimator in **every** city (Delhi 0.59, against 0.77 for hard fixed effects and 1.21 for the weighted design). For the pooled ATE it is best in Delhi and Kolkata, while posterior-feature DML is best in Mumbai and Bengaluru: scalar pooling avoids estimating weak regime contrasts. OR-DML's advantage here concerns *heterogeneous regime effects*, not universal scalar-ATE dominance. The weighted design is worse than standard DML outside Delhi.

The plug-in law, computed from the posterior, the fitted treatment shift and the injected outcome shift, ranks the four cities' biases correctly with magnitudes within 7–37%. The generated-state $\lambda_{\min}$ exceeds its oracle value by 1.7–2.4 times in Mumbai, Bengaluru and Kolkata (1.04 times in Delhi): phantom resolution is present in real-data-calibrated series too.

7.5 Representation quality is not task reliability

Across 15 shifted worlds and four encoders (HMM, GRU, Transformer, linear state-space model), adding posterior negative log-likelihood to state-recovery F_1 raises leave-one-world-out prediction of downstream reliability from $R^2 = 0.113$ to 0.221. Temperature scaling reduces calibration error of the GRU and Transformer by 47.9% and 60.3%, yet does not improve failure discrimination (ΔAUC -0.0035 and -0.0075). By Theorem 4, recalibration changes the *estimate* $\hat{v}$ but not the residual information that the downstream task actually has. A better-calibrated encoder is not a more reliable causal input.

7.6 Spectral regularization

Regularization trades bias for variance. On the committed path (Table 7.3) increasing λ from 0 to 0.1 cuts variance fivefold, from 0.0687 to 0.0143, and mean-squared error from 0.785 to 0.014; beyond $\lambda = 0.1$ bias returns. The best penalty is data dependent, which is why the implementation exposes an adaptive optimizer with an oracle risk bound.

Table 7.3. Spectral regularization path (λ against bias, variance and mean-squared error).

λ	$\lvert\text{bias}\rvert$	variance	MSE
0	0.8504	0.0687	0.7854
0.0001	0.8465	0.0684	0.7785
0.001	0.8131	0.0663	0.7210
0.005	0.6907	0.0583	0.5293
0.01	0.5784	0.0506	0.3795
0.02	0.4287	0.0395	0.2177
0.05	0.2122	0.0233	0.0615
0.1	0.0903	0.0143	0.0143
0.2	0.1259	0.0090	0.0209
0.5	0.3237	0.0048	0.1098

7.7 Scalability in the number of regimes

Table 7.4. Scalability check for $K = 2$ to 5 regimes: ATE bias, regime-effect error, smallest eigenvalue and condition number.

K	ATE bias	$\lVert\hat{\theta} - \theta\rVert_2$	$\lambda_{\min}$	κ
2	0.062	0.112	0.758	1.31
3	0.074	0.142	0.506	1.22
4	0.036	0.193	0.361	1.43
5	0.061	0.389	0.238	2.00

Regime-effect error grows with K (0.112 at $K = 2$ to 0.389 at $K = 5$) as $\lambda_{\min}$ falls (0.758 to 0.238), consistent with Theorem 3; ATE bias stays below 0.08.

7.8 What the synthetic evidence does not show

Below $\Delta_Z \approx 1$ every proxy-based estimator, OR-DML included, has large bias and no coverage. Posterior-adjusted OR-DML does not remove residual-state bias; it is the most accurate *regime-aware* estimator in the benchmarks reported, with ties at high separation. These limits are restated in Chapter 11.

CHAPTER 8

Observational Stress Test: Four Indian Cities

8.1 Purpose and framing

The stress test applies OR-DML and its sensitivity analysis to $NO_2 \rightarrow PM_{2.5}$ in Delhi, Mumbai, Bengaluru and Kolkata (OpenAQ/CPCB stations and Open-Meteo weather, February 2025–June 2026) on the corrected hourly grid. There is **no known truth** here. All numbers are *exposure associations*, and the purpose is to show what a careful analysis can and cannot conclude from real sensors. The framing is a stress test of the reporting rules developed in Chapter 5.

8.2 Results

Table 8.1. Four-city stress test on the corrected hourly grid (OR-DML, HAC lag 24 h). $\lambda_{\min}$ is raw, $\lambda_{\min}^{\text{std}}$ is scale-free. Lead placebo: smallest p-value over leads $h \in \{-6, -3, -1\}$ h. b^*: robustness value in SD of $\tilde{Y}$. Screened: runs of ≥ 6 identical NO_2 readings removed.

City	N	$\lambda_{\min}$	$\lambda_{\min}^{\text{std}}$	$\hat{\theta}_{\text{ATE}}$ (SE)	$\hat{\theta}_1$ (SE)	$\hat{\theta}_2$ (SE)	lead placebo	b^* (SD)
Delhi	9,024	63.2	0.46	0.234 (0.029)	0.430 (0.053)	0.102 (0.033)	rejects ($h=-1$, $p=0.012$)	82.3
Mumbai	8,113	0.0851	0.41	3.017 (1.682)	5.480 (2.416)	1.189 (2.429)	passes ($p_{\min}=0.33$)	11.9
Bengaluru	7,775	2.4	0.39	0.004 (0.074)	0.131 (0.140)	-0.132 (0.053)	rejects ($h=-6$, $p=0.016$)	26.8
Kolkata	2,322	0.00116	0.40	1.217 (1.143)	-0.674 (0.455)	3.894 (2.808)	passes ($p_{\min}=0.10$)	10.9
Mumbai (screened)	6,256	0.0495	0.20	0.383 (1.945)	-4.012 (4.687)	2.757 (1.496)	rejects ($h=-6$, $p=0.047$)	1.2
Bengaluru (screened)	7,180	2.3	0.40	0.020 (0.082)	0.185 (0.155)	-0.156 (0.062)	rejects ($h=-3$, $p=0.015$)	35.3
Kolkata (screened)	1,422	0.00172	0.40	3.319 (1.397)	0.191 (0.541)	6.916 (2.981)	rejects ($h=-1$, $p=0.000$)	16.5

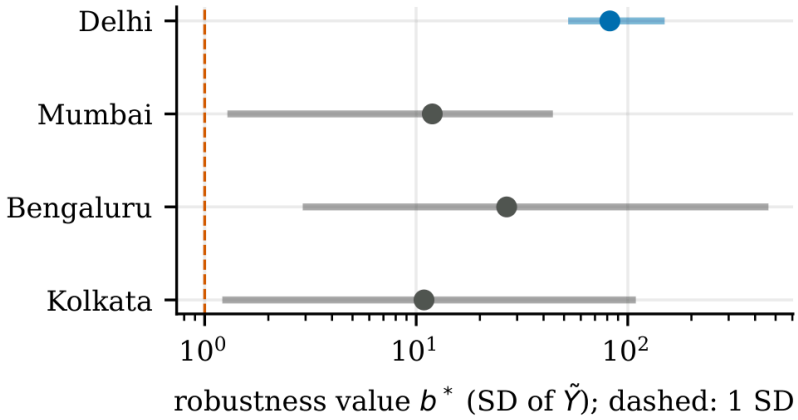


Figure 8.1. Robustness values b^* by city (log scale; dashed line = 1 SD). Delhi's association would need an outcome regime shift of 82 SD (90% interval 54–144) to be explained away through the HMM-regime channel; the other three cities are weak.

Table 8.1 separates four axes that the reliability rule conflates.

Scale. Raw $\lambda_{\min}$ ranges from 0.0012 to 63.2, but the standardized value only from 0.20 to 0.46. Raw conditioning reflects units and sensor resolution, not collinearity.

Data quality. Kolkata's NO_2 takes only 16 distinct values, and Mumbai's longest constant run is 277 hours; screening all qualifying runs moves the Mumbai ATE from 3.02 to 0.38 (N 8,113 $\rightarrow$ 6,256). The screen removes 1,857 complete cases (22.9%) across all flagged runs, not one episode. This is a sensitivity exclusion; it does not establish hardware failure or validate the remaining calibration.

Temporal validity. Bengaluru and screened Kolkata fail lead placebos, the latter strongly ($p = 0.0003$), so the apparently significant screened Kolkata effect (3.32) is *not interpretable*.

Residual confounding. Delhi's association is positive in both regimes (0.430 and 0.102; ATE 0.234, SE 0.029) and would need an outcome shift of 82 SD to be explained away through the HMM-regime channel. Its one-hour lead placebo nonetheless rejects (coefficient 0.072, $p = 0.012$, about a third of the contemporaneous coefficient), pointing to short-horizon confounding outside that channel.

8.3 Placebo falsification, aligned to the information set

Table 8.2. Lead and lag placebo coefficients (asterisk: $p < 0.05$). Leads $h < 0$ should be indistinguishable from zero if the temporal ordering is causal.

City	screen	$h=-6$	$h=-3$	$h=-1$	$h=0$	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$
Delhi	no	-0.02	0.02	0.07*	0.23*	0.70*	1.49*	1.61*	0.81*	0.33*
Mumbai	no	-0.35	-0.47	1.79	3.02	5.68*	2.73	-5.46*	-4.94	-9.90*
Bengaluru	no	-0.03*	-0.04*	-0.05	0.00	0.01	-0.01	0.01	-0.05	-0.10
Kolkata	no	-0.04	-0.17	0.91	1.22	1.05	-0.82	-2.71	-2.24	0.30
Delhi	yes	-0.02	0.02	0.07*	0.23*	0.70*	1.49*	1.61*	0.81*	0.33*
Mumbai	yes	-9.28*	3.16	5.42	0.38	4.17	-3.47	-4.52	-7.70	-15.61*
Bengaluru	yes	-0.04*	-0.05*	-0.06	0.02	0.03	0.00	-0.01	-0.08	-0.13*
Kolkata	yes	0.39	0.92	3.16*	3.32*	4.14*	1.77	3.88	17.20*	36.04*

Placebo leads are aligned to the outcome's own information set. Without this alignment the one-hour lead is mechanically zero, because the lagged outcome is a control; earlier drafts had that flaw (Chapter 10). In Delhi the lead coefficients at -6, -3 h are near zero but -1 h rejects; in Bengaluru the -6 and -3 h leads reject.

8.4 Sensitivity of inference to the HAC lag

Table 8.3. Regime effects (standard error; p-value) under HAC bandwidths of 12, 24, 72 and 168 hours.

City	regime	lag 12	lag 24	lag 72	lag 168
Delhi	1	0.430 (0.050; p=0.000)	0.430 (0.053; p=0.000)	0.430 (0.059; p=0.000)	0.430 (0.065; p=0.000)
	2	0.102 (0.033; p=0.002)	0.102 (0.033; p=0.002)	0.102 (0.036; p=0.005)	0.102 (0.040; p=0.011)
Mumbai	1	-4.012 (4.967; p=0.419)	-4.012 (4.687; p=0.392)	-4.012 (4.869; p=0.410)	-4.012 (5.223; p=0.442)
	2	2.757 (1.473; p=0.061)	2.757 (1.496; p=0.065)	2.757 (1.547; p=0.075)	2.757 (1.623; p=0.089)
Bengaluru	1	0.182 (0.149; p=0.222)	0.182 (0.155; p=0.240)	0.182 (0.183; p=0.320)	0.182 (0.196; p=0.354)
	2	-0.158 (0.059; p=0.008)	-0.158 (0.063; p=0.012)	-0.158 (0.073; p=0.031)	-0.158 (0.085; p=0.063)
Kolkata	1	-0.027 (0.522; p=0.959)	-0.027 (0.544; p=0.961)	-0.027 (0.617; p=0.965)	-0.027 (0.581; p=0.963)
	2	7.084 (3.450; p=0.040)	7.084 (3.401; p=0.037)	7.084 (3.279; p=0.031)	7.084 (3.260; p=0.030)

Delhi's regime-1 effect (0.430) stays significant at every bandwidth up to one week (standard error 0.050 $\rightarrow$ 0.065), as does regime 2 (0.102, p from 0.002 to 0.011). Kolkata's regime-2 effect is significant at every lag but rests on 2,322 observations and 16 distinct NO_2 values, so it should not be over-read. Mumbai is never significant.

8.5 Reading the results

The honest report is short.

- **Delhi** is an exposure association, positive in both regimes, robust to residual *regime* confounding. It is not an emissions effect: the lead placebo rejects.
- **Mumbai, Bengaluru, Kolkata** are inconclusive: data quality, failed placebos or wide intervals prevent interpretation.

NO_2 is also chemically related to the formation of secondary $PM_{2.5}$, proxies overlap controls, and lagged pollutants may be mediators. Chemistry-aware or instrument-based designs would be needed for emission-policy claims. A further point recorded in the manuscript is that five EM restarts span 1,182 log-likelihood units in Mumbai; neither optimizer multimodality nor misspecification establishes the direction of error in $\hat{v}$ or $b^{\star}$.

8.6 Why this section matters for the project

An applied study that had reported these four numbers as "the causal effect of traffic-related NO_2 on $PM_{2.5}$" would have been wrong in three of four cities and incomplete in the fourth. The project's contribution is partly the refusal to do so, and the tooling (aligned placebos, screening, robustness values, standardized conditioning) that makes the refusal principled rather than ad hoc.

CHAPTER 9

Engineering, Reproducibility and Verification

9.1 Repository architecture

The project is a research codebase of 496 commits (23 March 2026 to 7 October 2026). Its structure is designed so that a reviewer can move from a sentence in the paper to the file that produced it.

Table 9.1. Repository layout

Path	Contents
`paper/`	`main.tex`, appendix sources, checklist, bibliography; `generated/` holds `numbers.tex`, `claims.csv` and all `tab_*.tex`
`src/or_dml.py`	the OR-DML estimator (HMM, purged folds, coupled score, spectral solve, HAC)
`src/synthesis/`	consolidation experiments: closed form, orthogonality and coupled-law checks, frontier, learned posteriors, semi-synthetic, city stress test, figures, tables
`src/bias_law/`	residual-state bias-law experiments and real-data sensitivity analysis
`src/*.py`	factorial, frontier, representation zoo, scalability, soft-versus-hard, legacy analyses
`reports/`	all result files (CSV/JSON)
`data/raw_hourly/`	raw OpenAQ/CPCB and Open-Meteo hourly files
`archive/`	earlier manuscript lineages, legacy verifiers and drafts, kept for transparency
`research/`	research ledger (phantom resolution, directional reliability, and others)

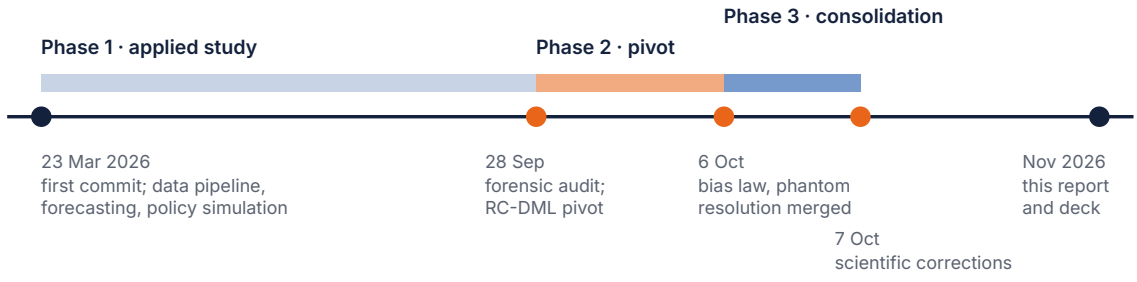


Figure 9.1. Project timeline. Three phases of the project (Chapter 10) on a calendar axis, from the first commit to this report.

9.2 A single source for every number

Every numeric claim in the manuscript's main text is a LaTeX macro defined in `paper/generated/numbers.tex` (247 macros), and every results table is an `\input` of `paper/generated/tab_*.tex`. Both are written *only* by `src/synthesis/make_tables_numbers.py` from committed CSV/JSON outputs. The ledger `paper/generated/claims.csv` maps each macro to its value, source file and derivation. As an example, the headline Spearman correlation is traceable as follows.

```
macro,value,source,how
FacRhoD,0.995,reports/factorial_reliability_correlations.csv,Spearman_Run_Difficulty
```

This design eliminates the most common source of error in research writing: a number typed into a paper that no longer matches the experiment. The numbers quoted in this report were taken from the same files.

9.3 The verifier

`verify_submission.py` performs five checks, and `python verify_submission.py --fast` runs them in minutes.

1. Regenerate all macros and tables into a temporary directory and require *byte-identical* output.
2. Re-evaluate the closed form and its symbolic identities.
3. Check that every macro used in the manuscript is defined and every defined macro is used or listed.
4. Check that the PDF is newer than its sources and that the main body is eight pages.
5. Scan the PDF and the supplementary archive for de-anonymizing strings (the submission is double-blind).

`verify_science.py` independently recomputes the analytical identities and the affected frontier, semi-synthetic, city and HAC-sensitivity results.

9.4 One-command reproduction

```
pip install -r requirements.txt
python verify_submission.py --fast   # PDF numbers vs result files
bash reproduce.sh                    # core experiments + rebuild (~12 min)
bash reproduce.sh --full             # plus long legacy benchmarks
```

`reproduce.sh` reruns the synthesis experiments (about ten minutes on two CPU cores), the factorial experiment (about 20 seconds), regenerates figures, tables and `numbers.tex`, and then runs the verifier. Two GitHub Actions workflows, `submission.yml` and

`scientific-reproduction.yml`, rerun the verifier and the LaTeX build on every push; neither commits to the repository.

9.5 Architecture decision records

Seven architecture decision records in `docs/adr/` document why the project's direction changed, including the methodological pivot, the RC-DML formulation, the overlap-aware pivot, the empirical diagnostics and figure standards, the oracle score-difference audit, the editorial compression of the paper, and the repository purge that consolidated the project's scientific identity. They are summarized in Chapter 10.

9.6 Supplementary material

The anonymized supplementary archive (`AISTATS2027_Supplementary_Material.zip`) contains the code, data, every result file, the LaTeX sources and the verifier, so that a reviewer can regenerate the paper from the archive alone.

CHAPTER 10

Project Journey: From Applied Study to a Research Contribution

A doctoral-style report is expected to say not only what was found but how the finding was reached and what was discarded on the way. This chapter does so. The project passed through three phases, and the audits at the phase boundaries are as much a result as the final theorems.

10.1 Phase 1: the applied air-quality study (March–September 2026)

The project began as an applied spatio-temporal air-quality informatics study across Indian cities. The codebase of this phase (preserved in `archive/pre-pivot/`) assembled hourly OpenAQ/CPCB pollution and Open-Meteo weather data, explored seasonal and spatial structure, and benchmarked forecasting models: LSTM and CNN-LSTM hybrids, a Temporal Fusion Transformer, random forests, LightGBM and stacking ensembles. It then attempted policy simulation (emission-reduction scenarios) and drafted a long manuscript aimed at a multi-disciplinary environmental journal.

The phase produced real engineering value: a robust ingestion and cleaning pipeline, a model benchmarking harness, and a clear empirical finding that tree ensembles matched or beat the deep sequence models on lagged meteorological tables. It also produced the claims that did not survive scrutiny.

10.2 The audit

Before submission, a forensic audit of the draft, performed by the team the way a skeptical reviewer would read it, found four problems serious enough to make the manuscript unpublishable at a top machine-learning venue.

Table 10.1. Findings of the forensic audit of the applied draft

Finding	What was wrong	Consequence
Decorated arithmetic	Health and economic benefits were computed by multiplying a pollution change by an arbitrary scalar, with a code comment that contradicted the constant used	headline economic claims had no derivation
A sign-flipping `abs()`	In Mumbai a naive model estimated a *negative* effect; taking the absolute value turned a worsening into thousands of "lives saved"	the policy simulation was invalid
Internal inconsistencies	MSE and RMSE in one table were reported in different units; a README cited a model score that was absent from the results; stage directions from an AI draft survived in the text	credibility of the whole table set
Baseline paradox	The paper advertised deep hybrids although its own benchmark showed tree ensembles winning	central claim undercut

The most instructive item is the first. A downstream "causal" number computed as pollution change times a scalar is not causal inference; it is arithmetic wearing a causal label. The remedy was not to patch the scalar but to retire every health and economic figure and to ask a harder question: *what does it take to estimate a causal effect from this kind of data at all?*

10.3 Phase 2: the methodological pivot (late September 2026)

The audit redirected the project towards a genuine open problem in causal machine learning: environmental time series violate standard DML in two ways, **latent non-stationary confounding** and **temporal autocorrelation** in cross-fitting. Phase 2 formulated *Regime-Conditional DML* (RC-DML): a coupled, multi-regime orthogonal score solved by inverting the full $K \times K$ Jacobian rather than regime-by-regime, purged block cross-fitting, and a variance decomposition for stationary averages. It produced the first versions of the omitted-regime bias decomposition, the non-identification result and the coupled-score theory used in this report, the first synthetic benchmarks (naive OLS bias of +525% against near-zero for RC-DML in the early design), and a real-data evaluation across three cities.

The pivot was documented as an architecture decision record, and the repository was restructured around the new contribution; the 30-page narrative was discarded.

10.4 Phase 3: consolidation and correction (October 2026)

Phase 3 is where most of the intellectual progress of the project occurred, and it came from *trying to break its own results*. Several development lines were merged on `master`: the submitted draft (which supplied the registered abstract and estimator theory), a residual-state bias-law analysis with a time-grid audit, and a research ledger on task-relative reliability (phantom resolution, directional reliability). Audits of the merged work changed the claims as follows.

Table 10.2. Claims that changed during consolidation

#	Earlier claim or artifact	Audit finding	Resolution
1	"Hourly" real-data lags	the panel sat on a two-hour grid	all real-data numbers recomputed (Section 6.3)
2	Placebo leads	not aligned to the outcome's information set; one-hour lead mechanically zero	aligned placebos
3	$\varepsilon_\gamma/\lambda_{\min}$ as a general reliability predictor	the factorial's 0.995 is algebraic; learned posteriors give 0.517	Proposition 1 and Corollary 1
4	Posterior-*weighted* nuisances as the method of choice	hard fixed effects beat it in every benchmark	posterior-*adjusted* design; weighted design kept as ablation
5	Semi-synthetic benchmark with the true state a function of the proxies	circular	replaced by real-data-calibrated benchmark with sampled states
6	Financial transfer, imputation sensitivity, multi-horizon forecasting tables, policy simulation	built on the legacy grid or unsupported	retired; code kept in the archive
7	Regularization numbers	stale	recomputed from the committed path

Scientific corrections after external feedback (7 October 2026)

A second round of review identified errors of mathematics and of claim scope, summarized below. Each was reproduced against the earlier implementation before it was fixed.

- **Non-identification proof.** The earlier construction did not satisfy proxy sufficiency. The replacement uses independent randomized treatment, independent memoryless states and uninformative proxies, with identical observable laws and different regime effects (Theorem 2 as stated here).
- **Inference appendix.** Rewritten for posterior-adjusted coupled moments under explicit sufficient conditions (bounded clipped nuisances, an absolutely regular augmented process, separated calibration and nuisance samples); full-sample smoothing is excluded.
- **HAC variance.** The canonical HAC now uses the moments actually solved, including off-diagonal terms, with scores centered at their regularized proxy target; the bootstrap uses the same moment matrices. Legacy uncertainty claims from the incorrect implementation were retired.
- **Embargo.** A search cap in the ACF routine could truncate the logarithmic floor; this was fixed and tested.
- **Bartlett bands** were reclassified as finite-lag operational diagnostics, not an estimate of the mixing exponent.

- **Bibliography.** Six references were checked against primary sources and corrected (titles, authors, arXiv identifiers).
- **Feedback assessed and partly rejected.** A suggestion that misspecification implies an optimistic lower bound was rejected as a universal claim (the posterior's Gini can err in either direction). The Mumbai restart instability motivated a footnote, not a theorem.

Five numerical regression tests were added, and the verifier now checks code–manuscript freshness.

10.5 What we learned

1. **Verification beats persuasion.** Every real improvement in the project was the result of an attempt to falsify it, whether a grid audit, an aligned placebo or an exact law.
2. **A good diagnostic is an object to be tested, not trusted.** The most widely used reliability rule was shown to fail in a precisely characterized way.
3. **Simple baselines are mandatory.** Hard fixed effects and posterior-feature DML were the competitors that forced the design change; the project's first estimator lost to them.
4. **State the scope of every result.** Delhi is an association, three other cities are inconclusive, and every estimator fails at weak separation. These sentences cost nothing and protect the whole.
5. **Single sources of truth for numbers** remove an entire class of error, and the cost is a few hundred lines of generating code.

10.6 Research output: the AISTATS conference

The research output required of this minor project is a peer-reviewed submission. Ours is the manuscript *Reliable Causal Estimation under Latent Markov Confounding*, submitted to the 30th **International Conference on Artificial Intelligence and Statistics (AISTATS 2027)**. This section explains what that venue is and how it is rated, because in computer science and artificial intelligence the standing of a conference differs from the convention in electrical and other engineering disciplines, where journals are the usual benchmark.

Conferences as primary publication venues. In machine learning and statistics, the top conferences are full-length, archival, peer-reviewed publication venues, not presentation forums. A complete paper is submitted, assessed in a double-blind review (typically three or more independent reviewers, then meta-review and an author rebuttal), and the accepted papers are printed in the conference proceedings, which are indexed in bibliographic databases such as DBLP and Google Scholar. The pace of the field is the reason: a journal review cycle of a year or more is too slow for it, and the leading results appear first at conferences.

What AISTATS is. AISTATS began as a small workshop series on artificial intelligence and statistics in Princeton, USA, in 1985, and grew into an annual conference. It is organised by the research community in the area where machine learning meets statistics. Its proceedings are published in the *Proceedings of Machine Learning Research* (PMLR), the series associated with the *Journal of Machine Learning Research*. The venue is known for rewarding theoretically grounded work with careful empirical validation, which is the character of this project.

Table 10.3. How AISTATS is rated and evaluated

Measure	What it means	AISTATS
CORE rank (Computing Research and Education Association of Australasia, ICORE portal)	Community-run classification of conferences: A* (flagship), A (excellent), B (good), C (other)	**A**, in the 2017, 2018, 2020, 2021 and 2023 rankings
Acceptance rate	Share of submissions accepted; a lower share means a more selective review	about **31%** in 2025 (583 of 1,861 submissions); about 29% in 2015; historically 25-31%
Submission volume	Size of the competing field	roughly 1,900 full-paper submissions per year
h5-index (Google Scholar Metrics)	Largest number *h* such that *h* papers from the last five complete years each have at least *h* citations; the standard citation-based venue score	101 (Google Scholar Metrics, as read by the authors; five-year window of 2020 to 2024)
Publication record	Where accepted papers appear and are indexed	PMLR; indexed by DBLP and Google Scholar

Reading these numbers. CORE A places AISTATS in the second tier of the four-tier scheme, directly below the handful of flagship conferences (such as NeurIPS, ICML and ICLR), and it is commonly treated as comparable in standing to a good journal in the field. An acceptance rate near 30% means that roughly seven of ten submissions, many of them strong, are rejected. The h5-index is the venue-level analogue of a journal's impact factor, and a value of 101 means that its top 101 papers of the last five years each have at least 100 citations.

Status of our submission. The manuscript, with its eight-page main body, appendices A-Q, and a reproducible supplementary archive, has been submitted to AISTATS 2027 and is under peer review. Its title and abstract match the registered submission, which the verifier checks. Acceptance is not asserted anywhere in this report; the outcome will be decided by the reviewers. The work was therefore formulated to the standard of a peer-reviewed venue of this rank and has entered its review process.

CHAPTER 11

Conclusions, Limitations and Future Work

11.1 Conclusions

The intuition that motivated the project was that a confident latent state together with a well-conditioned score Jacobian certifies a causal estimate. It does not. A generated state enters the downstream estimating equation twice, in its bias and in its apparent information, and the two move together. The scalar $\varepsilon_\gamma/\lambda_{\min}$ is an honest worst-case bound and an excellent ranking device when the state error is synthetic and the regime shifts are fixed. It degrades sharply once posteriors are learned (Spearman 0.995 to 0.517) because its denominator is partly made of the error that it is meant to measure.

Two rules survive.

- **Design.** Condition the nuisances on the posterior and use the posterior only to weight the target. This makes the coupled score exactly orthogonal and gave the most accurate regime estimates in every benchmark of the project.
- **Reporting.** State the residual state variance, the regime shifts it multiplies and a robustness value; compare against hard fixed-effects and posterior-feature baselines; and test temporal validity with aligned placebos.

Against the objectives of Chapter 1: the dataset was assembled and audited (O1); the propagation of state error was characterized exactly (O2); an estimator with proven orthogonality was implemented and verified (O3); it was validated from fully synthetic to semi-synthetic to real data (O4); a peer-reviewed research submission was produced (O5); and every number is regenerable and machine-checked (O6).

11.2 Limitations

- The laws are probability limits under linear regime shifts, exogenous treatment innovations and calibrated posteriors.
- The robustness value bounds rather than corrects, and it depends on calibration under misspecified emissions.
- Under weak observability every proxy-based estimator, OR-DML included, has large bias and no coverage. For the pooled ATE, posterior-feature DML is sometimes more accurate.
- The four-city analysis is observational: proxies overlap controls, lagged pollutants may be mediators, NO_2 is chemically related to $PM_{2.5}$, and Delhi's lead placebo rejects.

- Inference theory covers posterior-adjusted coupled moments and excludes full-sample smoothing; implemented intervals condition on the fitted HMM.
- The raw data span about sixteen months, and Kolkata's about four, which limits seasonal generalization.

11.3 Threats to validity

- **Construct validity.** NO_2 stands in for traffic and combustion emissions, and $PM_{2.5}$ for particulate burden. Neither is a policy intervention, and NO_2 takes part in the chemistry that forms secondary $PM_{2.5}$. The estimated quantity is therefore an exposure association unless a policy-shock design supplies exogenous variation.
- **Internal validity.** The simulation designs use linear regime shifts and Gaussian proxies, so the exact laws are exact for those designs only. The real-data-calibrated benchmark relaxes this by sampling states from an HMM fitted to each city, but the injected causal effects are still known constants. Most experiments use two regimes, which could favour a two-state method; the continuous-mixture experiment (Section 7.3) and the scalability check to $K = 5$ (Section 7.7) address this only in part.
- **Statistical conclusion validity.** Replication counts are 30 to 100 datasets per cell. Differences between estimators that are small relative to Monte Carlo error (for example at high proxy separation) are reported as ties, not wins. Observational intervals condition on the fitted HMM and ignore its estimation uncertainty.
- **External validity.** The data cover about sixteen months and four cities (about four months for Kolkata), with sensors of uneven quality. Seasonal and multi-year generalization is untested, and conclusions about specific cities should not be transferred to other airsheds.
- **Reproducibility risk.** Every number is regenerated from committed files, but the raw sensor files are third-party data whose upstream records may be revised.

11.4 Future work

The dual-factor view of reliability. Preliminary experiments (documented in `docs/FUTURE_WORK_AND_FRONTIERS.md`) suggest that failure is predicted by the *product* of representation ambiguity and task conditioning, rather than by either factor alone, and that selective abstention on the hardest windows can reduce risk sharply. These results predate the corrections of Chapter 10 and are therefore proposed as hypotheses to be re-tested under the learned-posterior protocol, not as findings.

Software. Packaging the estimator as an open-source library (`py-ordml`) with a scikit-learn-style interface, modular regime models, EconML compatibility and interactive documentation.

Theory. Extension beyond first-order Markov chains to semi-Markov dwell times; end-to-end training of the state model with a causal loss while preserving orthogonality; distribution-free (conformal) guarantees for abstention.

Domains. The mechanism is not specific to HMMs. Any generated state, such as a neural encoder's regime embedding, a clustering label or a changepoint segmentation, enters a downstream estimating equation both as an adjustment and as a source of apparent information. Applications that share the structure include intensive-care treatment under latent physiological phases, execution-cost estimation under latent market regimes, and offline reinforcement learning in partially observable environments.

Air quality. Chemistry-aware treatment definitions and instrument-based designs (for example, policy shocks such as lockdowns or festival-related restrictions) are the natural route to emission-policy claims that the present observational analysis cannot make.

11.5 Closing remark

The project began by trying to estimate how much traffic-related pollution matters. It ended by establishing, exactly and checkably, when such an estimate from latent-state time series should and should not be believed, and by building the tools to report that distinction. In a field where confident numbers are cheap, this is the contribution we are most willing to defend.

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ANNEXURE A: TEAM AND PROJECT ADMINISTRATION

Table A.1. Project team and supervision

Name	Registration No.	School	Role
Pulkit Kumar	230906192	Electrical Engineering	team member
Utsav Verma	230906526	Electrical Engineering	team member
Nishant Kumar Upadhyay	230906522	Electrical Engineering	team member
Dr. Savitha G	n/a	Computer Engineering	project guide

Programme. Minor in Data Science, 8 credits, 7th semester, academic year 2026-27, towards the Bachelor of Technology degree in Electrical and Electronics Engineering of Manipal Institute of Technology. **Required output.** A peer-reviewed research submission (the AISTATS 2027 manuscript).

ANNEXURE B: AI-USE STATEMENT

Large language models assisted with code, derivations, experiment scripting and drafting. Every number in the manuscript is generated from committed artifacts by a single script (`src/synthesis/make_tables_numbers.py`) and checked by `verify_submission.py`, which re-derives all tables and macros and fails on any mismatch. The closed forms are verified symbolically, and the errors found in audit in earlier drafts (time grid, placebo alignment, overstated estimator claims) are documented in Chapter 10 and in Appendix Q of the manuscript. The authors reviewed all arguments and take full responsibility for the content. This report was also prepared with AI assistance; its numbers are drawn from the same generated files and its figures from the committed plots.

ANNEXURE C: KEY CODE LISTINGS

C.1 Embargo selection (`src/or_dml.py`). Finite-lag autocorrelation bands choose the embargo, floored at $C \log N$.

```
def estimate_optimal_embargo(X, Z, max_tau=24, c_log=2.0):
    N = len(X); crit = 1.96 / np.sqrt(N)
    V = np.column_stack([X, Z]); V_centered = V - np.mean(V, axis=0)
    var_V = np.var(V, axis=0) + 1e-12
    search_tau = min(max_tau, max(12, int(N // 10)))
    significant_lags = [0]
    for h in range(1, search_tau + 1):
        cov_h = np.mean(V_centered[:-h] * V_centered[h:], axis=0)
        if np.any(np.abs(cov_h / var_V) ⩾ crit):
            significant_lags.append(h)
    bartlett_tau = max(significant_lags) + 1
    min_log_tau = int(np.ceil(c_log * np.log(max(N, 2))))
    return int(max(bartlett_tau, min_log_tau, 1))
```

C.2 Purged block cross-validation.

```
class PurgedBlockKFold:
    def split(self, N):
        indices = np.arange(N)
        sizes = np.full(self.n_splits, N // self.n_splits, dtype=int)
        sizes[:N % self.n_splits] += 1
        current = 0
        for size in sizes:
            test_idx = indices[current:current + size]
            train_mask = np.ones(N, dtype=bool)
            lo = max(0, current - self.embargo_tau)
            hi = min(N, current + size + self.embargo_tau)
            train_mask[lo:hi] = False          # purge + embargo on both sides
            yield indices[train_mask], test_idx
            current += size
```

C.3 Coupled solve with spectral penalty (condensed from `OverlapAwareRegimeDML.fit`).

```
eigvals = np.linalg.eigvalsh(J_mat)           # diagnose conditioning first
try:
    inv_J_reg = np.linalg.inv(J_mat + eff_lambda * np.eye(K))
except np.linalg.LinAlgError:
    inv_J_reg = np.linalg.pinv(J_mat + eff_lambda * np.eye(K))
theta_vec = inv_J_reg @ S_vec
```

C.4 Planned public API (`docs/FUTURE_WORK_AND_FRONTIERS.md` ; not yet released).

```
estimator = OverlapAwareRegimeDML(
    regime_model=GaussianHMMRegimeModel(n_regimes=2),
    model_y=LGBMRegressor(), model_t=LGBMRegressor(),
    spectral_penalty="adaptive_plugin", embargo_buffer="auto_bartlett", n_splits=5)
estimator.fit(X=X_env, T=treatment_t, Y=outcome_y, Z=proxy_z)
```

ANNEXURE D: REPRODUCTION COMMANDS

```
git clone <repository> && cd aqi-informatics-india
pip install -r requirements.txt           # or: conda env create -f environment.yml
python verify_submission.py --fast        # numbers, macros, page budget
bash reproduce.sh                         # ~12 min on 2 cores
bash reproduce.sh --full                  # also the long legacy benchmarks
```

Raw data are in `data/raw_hourly/` ; all result files in `reports/` ; generated artifacts in `paper/generated/` .

ANNEXURE E: ADDITIONAL RESULTS

Table E.1. Synthetic frontier, full estimator list: mean absolute ATE bias, median, RMSE and smallest eigenvalue at each proxy separation Δ_Z (100 datasets per row).

Δ_Z	Estimator	mean $\lvert\text{bias}\rvert$	median	RMSE	$\lambda_{\min}$
0.2	Oracle DML	0.056	0.044	0.070	0.449
	Standard DML	8.456	8.456	8.456	4.934
	Block DML	8.490	8.479	8.491	5.011
	DML + Z Controls (Ridge)	8.307	8.312	8.308	4.627
	DML + Z Controls (GBM)	8.018	8.025	8.019	5.325
	Regime FE DML (Hard)	8.318	8.324	8.319	1.745
	Unregularized Coupled DML	8.637	8.615	8.638	0.804
	Spectral OR-DML	8.627	8.604	8.628	0.804
	Filtered OR-DML	8.604	8.582	8.605	0.800
	Decoupled Soft OR-DML	8.381	8.381	8.381	1.534
	HMM(Z,T) OR-DML	6.504	6.606	6.546	0.487
0.5	Oracle DML	0.055	0.044	0.071	0.448
	Standard DML	8.429	8.432	8.430	4.956
	Block DML	8.467	8.483	8.468	5.041
	DML + Z Controls (Ridge)	7.429	7.424	7.430	3.401
	DML + Z Controls (GBM)	7.031	7.020	7.034	3.819
	Regime FE DML (Hard)	6.628	6.672	6.634	1.203
	Unregularized Coupled DML	8.848	8.887	8.851	0.828
	Spectral OR-DML	8.835	8.874	8.839	0.828
	Filtered OR-DML	8.988	9.003	8.991	0.875
	Decoupled Soft OR-DML	7.164	7.178	7.167	1.311
	HMM(Z,T) OR-DML	4.324	4.390	4.364	0.482
1.0	Oracle DML	0.057	0.043	0.073	0.452
	Standard DML	8.433	8.449	8.434	4.954
	Block DML	8.466	8.475	8.467	5.027
	DML + Z Controls (Ridge)	5.438	5.462	5.440	2.078
	DML + Z Controls (GBM)	3.633	3.642	3.647	1.885
	Regime FE DML (Hard)	2.647	2.700	2.685	0.609
	Unregularized Coupled DML	4.025	4.030	4.045	0.557
	Spectral OR-DML	4.017	4.022	4.037	0.557
	Filtered OR-DML	5.172	5.185	5.185	0.605
	Decoupled Soft OR-DML	3.294	3.286	3.310	0.670
	HMM(Z,T) OR-DML	1.196	1.192	1.230	0.460
2.0	Oracle DML	0.052	0.042	0.066	0.454
	Standard DML	8.432	8.451	8.433	4.954
	Block DML	8.461	8.472	8.462	5.024

	DML + Z Controls (Ridge)	2.631	2.614	2.635	1.351
	DML + Z Controls (GBM)	0.207	0.194	0.243	1.214
	Regime FE DML (Hard)	0.077	0.063	0.102	0.455
	Unregularized Coupled DML	0.082	0.073	0.105	0.455
	Spectral OR-DML	0.081	0.072	0.104	0.455
	Filtered OR-DML	0.102	0.081	0.131	0.456
	Decoupled Soft OR-DML	0.079	0.069	0.101	0.456
	HMM(Z,T) OR-DML	0.059	0.045	0.074	0.454
4.0	Oracle DML	0.054	0.048	0.067	0.439
	Standard DML	8.451	8.461	8.452	4.912
	Block DML	8.486	8.495	8.487	4.996
	DML + Z Controls (Ridge)	0.848	0.842	0.853	1.095
	DML + Z Controls (GBM)	0.101	0.082	0.125	1.189
	Regime FE DML (Hard)	0.054	0.048	0.067	0.439
	Unregularized Coupled DML	0.054	0.048	0.067	0.439
	Spectral OR-DML	0.054	0.046	0.067	0.439
	Filtered OR-DML	0.054	0.046	0.067	0.439
	Decoupled Soft OR-DML	0.054	0.046	0.067	0.439
	HMM(Z,T) OR-DML	0.054	0.046	0.067	0.439

ANNEXURE F: MAP OF THE MANUSCRIPT APPENDICES

The AISTATS manuscript's supplementary material is in appendices A–Q. They are the formal back-up for this report.

Table F.1. Manuscript appendices and where this report uses them

App.	Title	Used in
A	Assumptions	Chapter 3
B	Proof of Theorem 1 (omitted regime bias)	Section 3.2
C	Proof of Theorem 2 (non-identification)	Section 3.3
D	Proof of the graceful-degradation theorem	Section 5.2
E	Posterior-adjusted nuisances	Sections 4.2, 5.1
F	The coupled score under state confusion	Section 5.3
G	The residual-state bias law	Section 5.4
H	Inference	Section 5.5
I	Operational surrogate, screening and regularization	Sections 4.2, 7.6
J	Algorithm and scalability	Sections 4.2, 7.7
K	Factorial and learned-posterior experiments	Sections 7.1, 7.2
L	Synthetic frontiers	Section 7.3
M	Real-data-calibrated semi-synthetic benchmark	Section 7.4
N	Representations, calibration and reliability	Section 7.5
O	Sensor data, audit and full city results	Chapter 8, Section 6.3
P	Reproduction and verification	Chapter 9
Q	Lineage, corrections and scope	Chapter 10

ANNEXURE G: ASSUMPTIONS AND PROOFS

This annexure states the assumptions used in the theory and gives the complete derivations of the results that are proved by direct computation. The remaining proofs (graceful degradation, posterior-adjusted nuisances, asymptotics) are in appendices D, E and H of the manuscript (Annexure F).

G.1 Assumptions

(A1) Unconfoundedness and innovation exogeneity. Conditional on X_t, S_t and the proxy history $Z_{1:N}$, potential outcomes are independent of T_t, and $\mathbb{E}[U_t \mid X_t, S_t, T_t, Z_{1:N}] = 0$.

(A2) Overlap, positivity and envelopes. The stationary law satisfies $\min_k \pi_k^* \geq \eta > 0$, and $P(S_t = k \mid X_t = x) \geq c_0 > 0$ almost everywhere. Variables are bounded, $|T_t| \leq B_T$ and $|Y_t| \leq B_Y$. Within each regime the treatment variance is bounded away from zero and infinity, $c_1 \leq \mathrm{Var}(T_t \mid X_t, S_t = k) \leq c_2$.

(A3) Latent proxy sufficiency. $P(S_t = k \mid X_t, T_t, Z_{1:N}) = P(S_t = k \mid Z_{1:N}) = \gamma_{tk}$ (or the filtered version). This assumption is used for the structural interpretation of the posterior; the degradation bound of Theorem 3 does not require it, and the simulation designs deliberately violate it.

(A4) Mixing and nuisance rates. The process is strictly stationary and geometrically α-mixing, $\alpha(m) \leq C_\alpha e^{-c_\alpha m}$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2} \, \|\hat{\boldsymbol{\mu}}_D - \boldsymbol{\mu}_D\|_{L_2} = o_P(N^{-1/2})$, with individual rates $o_P(N^{-1/4})$.

G.2 Proof of Theorem 1 (omitted regime bias)

Let $\pi_k(X) = P(S_t{=}k \mid X_t)$, $m_k(X) = \mathbb{E}[T_t \mid X_t, S_t{=}k]$, $\bar{m}(X) = \sum_k \pi_k m_k$, $g_k(X) = g(X, k)$ and $\bar{g} = \sum_k \pi_k g_k$. Residualizing on X alone gives

$$\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_k 1(S_t{=}k)\, \Delta m_k(X_t) + V_t, \qquad \Delta m_k = m_k - \bar{m}.$$

The outcome residual is $\tilde{Y}_t = \sum_k \theta_k^* 1(S_t{=}k) T_t - \sum_k \theta_k^* \pi_k m_k + g(X_t, S_t) - \bar{g}(X_t) + U_t$. Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment term splits into $\sum_k \theta_k^* 1(S_t{=}k)\tilde{T}_t$ plus a term in $\bar{m}(X_t)1(S_t{=}k) - \pi_k m_k$. Now take the inner product with $\tilde{T}_t$ and the expectation, term by term.

1. $\mathbb{E}[\tilde{T}_t U_t] = 0$ by (A1).
2. $\mathbb{E}\big[\tilde{T}_t \sum_k \theta_k^* 1(S_t{=}k)\tilde{T}_t\big] = \mathbb{E}[\tilde{T}_t^2] \sum_k \omega_k \theta_k^*$ with $\omega_k = \mathbb{E}[1(S_t{=}k)\tilde{T}_t^2] / \mathbb{E}[\tilde{T}_t^2]$ and $\sum_k \omega_k = 1$.

3. Because $\mathbb{E}[\tilde{T}_t \mid X_t] = 0$ and $\mathbb{E}[\tilde{T}_t 1(S_t{=}k) \mid X_t] = \pi_k \Delta m_k$, the mean-adjustment term equals $\sum_k \theta_k^* \mathbb{E}[\bar{m}\, \Delta m_k\, \pi_k]$, which is $\mathcal{B}_{\text{het}}\, \mathbb{E}[\tilde{T}_t^2]$.
4. The confounding term is $\mathbb{E}[\tilde{T}_t(g(X_t, S_t) - \bar{g}(X_t))] = \sum_k \mathbb{E}[\pi_k \Delta m_k \Delta g_k]$, which is $\mathcal{B}_{\text{conf}}\, \mathbb{E}[\tilde{T}_t^2]$.

Dividing by $\mathbb{E}[\tilde{T}_t^2]$ gives the stated decomposition. □

G.3 Proof of Proposition 1 and Corollary 1

Setting. With oracle regime-wise nuisances, the regime-k residuals evaluated on an observation from regime s are $\tilde{T}_k = (m_s - m_k) + \sigma_s V$ and $\tilde{Y}_k = (\mu_s - \mu_k) + \theta_s \sigma_s V + U$, where V has mean zero and unit variance and is independent of U and of S. The covariate parts cancel because the nuisances are regime-correct. Since γ_j is a function of s alone,

$$J_{jk} = \sum_s \pi_s \gamma_j(s) \gamma_k(s) \big[(m_s - m_j)(m_s - m_k) + \sigma_s^2\big], \qquad S_j = \sum_s \pi_s \gamma_j(s) \big[(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2\big].$$

Symmetric case. Put $a = 1 - \alpha$, $b = \alpha$, $\pi_s = \tfrac{1}{2}$, $\sigma_s = \sigma$. The diagonal entries are $\tfrac{1}{2}[a^2\sigma^2 + b^2(\Delta^2 + \sigma^2)]$. The off-diagonal entry is $ab\sigma^2$, because on either regime the two residuals are σV and $\pm\Delta + \sigma V$, whose product has mean σ^2. A symmetric 2×2 matrix with equal diagonal has eigenvectors $(1, \pm 1)$ and eigenvalues "diagonal $\pm$ off-diagonal", which gives

$$\lambda_+ = \tfrac{1}{2}\sigma^2 + \tfrac{1}{2}\alpha^2\Delta^2, \qquad \lambda_- = \tfrac{1}{2}\sigma^2(1 - 2\alpha)^2 + \tfrac{1}{2}\alpha^2\Delta^2$$

after using $a + b = 1$. For the displacement write $S - J\theta^* = b_\gamma$, project b_γ on $(1, \pm 1)$ and divide by $2\lambda_\pm$. The $(1, 1)$ component collects the leaked outcome contrast $\alpha\Delta G$ and the quadratic leakage $-\tfrac{1}{2}\alpha^2\Delta^2(\theta_0 + \theta_1)$; the $(1, -1)$ component collects the effect heterogeneity. The identity is checked symbolically in `src/synthesis/closed_form_factorial.py`.

Corollary 1. (a) With $\alpha = \varepsilon_\gamma/2$, $\lambda_-(\varepsilon_\gamma) - \sigma^2/2 = \tfrac{1}{2}\sigma^2(\varepsilon_\gamma^2 - 2\varepsilon_\gamma) + \varepsilon_\gamma^2\Delta^2/8$, which is positive exactly when $\varepsilon_\gamma(\sigma^2 + \Delta^2/4) > 2\sigma^2$, that is $\varepsilon_\gamma > 8\sigma^2/(\Delta^2 + 4\sigma^2)$. (b) Dropping the $\mathcal{O}(\alpha^2)$ numerator term and the heterogeneity component, the error is $\approx \sqrt{2}\,\alpha|\Delta G|/(\sigma^2 + \alpha^2\Delta^2)$. Maximizing $\varepsilon_\gamma/(\sigma^2 + \varepsilon_\gamma^2\Delta^2/4)$ gives $\varepsilon^\star = 2\sigma/|\Delta|$ and the stated peak. (c) The leading displacement is proportional to $(1, 1)$, the λ_+ eigenvector, so $\|J^{-1}b_\gamma\| = \|b_\gamma\|/\lambda_+ \le \|b_\gamma\|/\lambda_-$. □

G.4 Proof of Theorem 4(b): pooled residual-state bias

Take homogeneous effects ($\theta_0 = \theta_1 = \theta$), treatment $T_t = m(X_t) + aS_t + V_t$ and outcome $Y_t = \theta T_t + g(X_t) + bS_t + U_t$. Because the posterior is calibrated given the features F_t, $\mathbb{E}[S_t \mid F_t] = \gamma_t$, so the posterior-adjusted nuisances are $\mathbb{E}[T \mid F] = m(X) + a\gamma$ and $\mathbb{E}[Y \mid F] = \theta(m + a\gamma) + g + b\gamma$. The residuals are

$$\tilde{T}_t = a(S_t - \gamma_t) + V_t, \qquad \tilde{Y}_t = \theta\tilde{T}_t + b(S_t - \gamma_t) + U_t.$$

Hence the pooled residual-on-residual estimator converges to

$$\theta + \frac{\mathbb{E}[\tilde{T}_t\, b(S_t - \gamma_t)]}{\mathbb{E}[\tilde{T}_t^2]} = \theta + \frac{ab\,\mathbb{E}[(S_t - \gamma_t)^2]}{a^2\mathbb{E}[(S_t - \gamma_t)^2] + \sigma^2} = \theta + \frac{ab\,v}{a^2 v + \sigma^2},$$

where calibration gives $\mathbb{E}[(S_t - \gamma_t)^2] = \mathbb{E}[\gamma_t(1 - \gamma_t)] = v$, and V_t is independent of S_t with variance σ^2. The bias is $b\,v\,h(a)$ with $h(a) = a/(a^2 v + \sigma^2)$. Setting $h'(a) = 0$ gives $a^2 v = \sigma^2$, so the peak is at $|a| = \sigma/\sqrt{v}$ with value $h = 1/(2\sigma\sqrt{v})$ and therefore $\sup|B| = |b|\sqrt{v}/(2\sigma)$. The bias is zero at $a = 0$ (statement (a)). Statement (c) follows because $J = \mathbb{E}[ww^\top(a^2 v + \sigma^2)]$ grows with $a^2 v$. □

G.5 Proof sketch of Lemma 1 and Theorem 2

Lemma 1 is proved in Section 5.1. *Theorem 2* is proved by the construction in Section 3.3: the two binary-outcome models in Table G.1 have identical laws of $(Y, T, X, Z)_{1:N}$ for every N, because $P(Y_t = 1 \mid T_t = 0) = 0.40$ and $P(Y_t = 1 \mid T_t = 1) = 0.60$ in both, while their regime-effect vectors differ.

Table G.1. The two models of the non-identification construction

Model	θ_0	θ_1	g_0	g_1	$P(Y{=}1 \mid T{=}0)$	$P(Y{=}1 \mid T{=}1)$
P_1	0.20	0.20	0.30	0.50	0.40	0.60
P_2	0.10	0.30	0.40	0.40	0.40	0.60

Both models have population ATE 0.20, which randomized treatment identifies. The result is non-identification of the *regime-specific vector* within the stated memoryless model class; it does not imply that every persistent latent model with uninformative proxies is unidentified.